Measuring Shortages since 1900*

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Abstract

This paper introduces a monthly shortage index spanning 1900 to the present, constructed from 25 million newspaper articles. The index captures shortages across industry, labor, food, and energy, and spikes during economic crises and wars. We validate the index and show that it contains information beyond traditional macroeconomic indicators. Using predictive regressions, we find that shortages are associated with persistently high inflation and lower economic activity. A structural VAR model reveals that, compared to traditional supply shocks, surprise movements in shortages produce less inflation relative to their GDP impact, indicating that shortages are associated with constraints on price adjustment that limit inflation but magnify the decline in real activity. We also show that post-pandemic shortages and inflation were primarily driven by supply-side forces, with demand factors playing a less important role.

KEYWORDS: Shortages; Inflation; Textual Analysis; Predictive Regressions; Structural VAR Model.

JEL CLASSIFICATION: C32, C55, E31, N10.

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1 Introduction

Shortages—defined as the inability of supply to meet demand at prevailing prices—have been a recurring feature of economic life throughout the 20th and early 21st centuries. From widespread disruptions during major wars to the acute shortages of goods and labor during the COVID-19 pandemic, these episodes have strained supply chains, hindered production, and placed significant burdens on households and businesses. Despite their importance, systematic evidence on the long-run behavior of shortages across sectors and their effects remains scarce. This paper addresses this gap by providing a comprehensive analysis of shortages over a 125-year period and demonstrating their persistent effects on inflation and economic activity.

In the first part, we construct a monthly shortage index based on newspaper articles spanning from 1900 through the present (Section 2). This index, covering industry, labor, food and energy shortages, is derived from approximately 25 million articles from six major U.S. newspapers, providing a novel, long-term perspective on global supply disruptions. The index captures historical events effectively, spikes during periods of economic turmoil such as the two world wars and the 1970s oil crises, and reaches its highest level in 40 years during the COVID-19 pandemic.

Although developed using U.S. news sources, our index captures shortages on a global scale. We demonstrate this property by constructing sub-indexes for six major advanced economies—Canada, France, Germany, Japan, the United Kingdom, and the United States. These indexes spike around common global events, but also reflect country-specific variation, capturing both differential exposure to global shocks and idiosyncratic episodes, such as the Suez Crisis in 1956 or the 2011 earthquake and tsunami in Japan. In Section 3, we validate our index through narrative investigation, text analysis based on artificial intelligence (AI), and comparison to alternative measures of supply-side disruptions.

The second part shows that shortages are associated with higher inflation and lower economic activity in the United States. Using predictive regressions (Section 4), we document that shortages predict lower future GDP and higher future prices. These results hold even when controlling for traditional indicators of inflation pressures such as commodity prices and wages, suggesting that the shortage index provides additional information not captured by conventional measures of price and wage pressures.

To investigate how the relationship between shortages and activity varies over time, we

estimate rolling regressions of future prices and future GDP on shortages. Throughout most historical periods, shortages have been associated with higher future inflation. The predictive power for future GDP growth is typically negative, with notable exceptions during World War II and the COVID-19 pandemic—periods when shortages coincided with strong demand pressures—suggesting that both supply and demand forces are important for understanding shortage dynamics.

The varying impact of shortages and their persistent effect on inflation motivate our structural vector autoregressive (VAR) model in Section 5. A structurally identified VAR model offers a comprehensive and economically interpretable framework for understanding joint movements in GDP, inflation, and shortages. Our identification strategy relies on two assumptions: first, demand and supply shocks are identified based on the slope of economy-wide demand and supply curves and can affect prices, activity, and shortages contemporaneously; second, shortages drive a time-varying wedge between aggregate demand and aggregate supply, that can vary either in response to traditional demand or supply disturbances, or in response to exogenous shortage shocks. We interpret shortage shocks as unexpected disruptions that require two concurrent conditions to materialize. First, there must be supply constraints stemming from events such as extreme weather, natural disasters, geopolitical conflicts, labor strikes, or production failures. Second, unusual market frictions must impede normal price adjustment processes. These frictions include hoarding behavior, regulatory interventions, social norms against price increases, or coordination failures. Together, these two conditions result in quantity rationing that persists beyond what traditional market clearing would predict, and differentiate shortage shocks from traditional adverse supply shocks. The combination of supply disruptions with market frictions that impede price adjustment can potentially result in larger quantity reductions relative to price increases, distinguishing shortage shocks from conventional supply shocks where prices bear relatively more of the adjustment burden.

The VAR results reveal several key insights. While traditional economic forces contribute to the identified movements in the shortage index, exogenous shortage shocks remain the dominant driving force behind these fluctuations. Shortage shocks have significant and persistent effects on GDP and inflation, showcasing the shortage index's value as a leading indicator of activity and prices. Compared to traditional contractionary supply shocks, shortage shocks produce less inflation relative to their impact on economic activity, suggesting that shortages often involve constraints on price adjustment that limit inflation but magnify the decline in real

activity.

The historical decompositions from the VAR provide new perspectives on major economic episodes. Shortages in the 1950s were largely driven by the Korean War, post-World War II pent-up demand, and the rapid economic transition from wartime to peacetime. The 1970s oil embargo triggered shortages beyond what could be explained by adverse supply shocks alone. During the COVID-19 pandemic, shortages developed from a confluence of demand, supply, and shortage-specific shocks. According to the model, the rise in shortages and inflation that started in 2020 was predominantly driven by supply forces—a combination of traditional supply shocks, shocks to commodity prices, and shocks to shortages—with demand forces playing a somewhat secondary role. The reversal of adverse supply shocks and negative demand shocks from 2022 onward led to a faster-than-expected decline in inflation.[1]

Our approach builds on previous work combining newspaper sources and textual analysis to measure economic phenomena, such as the Economic Policy Uncertainty Index (Baker, Bloom, and Davis, 2016) and the Geopolitical Risk Index (Caldara and Iacoviello, 2022). While other researchers have developed news-based shortage indicators for shorter periods (Lamont, 1997; Chen and Houle, 2023; Burriel et al., 2023) or constructed alternative supply disruption measures using transportation and shipping data (Benigno et al., 2022; Bai et al., 2024; Liu et al., 2024; Känzig and Raghavan, 2025), our contribution is distinct in three ways. First, we develop the first comprehensive shortage index covering more than 125 years, offering a unique long-term perspective unavailable in existing indicators. Second, our disaggregated indexes capture events in different markets at both global and country-specific levels, providing insights for historical and cross-country analysis. Third, our empirical analysis sheds light on causes and consequences of shortages, finding that shortages have more persistent effects on inflation than previously recognized. Unlike Bernanke and Blanchard (2023), who use Google searches for the term "shortage" as a proxy for supply disruptions and find only short-lived inflationary effects from shortages, our longer-term analysis reveals more prolonged inflationary responses.

The remainder of the paper is structured as follows. Section 2 discusses the construction of the index and its evolution over time. Section 3 validates the index. Sections 4 and 5 examine the relationship between the shortage index and U.S. economic outcomes using predictive

[1] Our analysis of the post-pandemic drivers of inflation relates to a large and growing literature. Giannone and Primiceri (2024) and Bergholt et al. (2024) find that the pandemic-era rise in inflation was mostly driven by demand factors. By contrast, Ascari et al. (2024), Comin et al. (2024), Shapiro (2024) and Di Giovanni et al. (2023) find a relatively larger role for supply factors.

regressions and a structural VAR model. Section 6 concludes.

2 The Shortage Index

In this section, we begin by discussing the construction of the shortage index. We then show how the index captures key episodes of shortages in U.S. history.

2.1 Construction of the Index

The shortage index is a monthly news-based indicator that measures the intensity of shortages in materials, goods, labor, and energy in the United States. It is constructed from a sample of approximately 20,000 news articles per month, spanning from 1900 through present—about 25 million articles in total. These articles are published in major U.S. newspapers: the *Boston Globe*, the *Chicago Tribune*, the *Los Angeles Times*, *The New York Times*, *The Wall Street Journal*, and *The Washington Post*. Each month, the index counts articles that discuss shortages in energy, food, industry or labor markets—the set $\mathcal{S}$ visualized in the top panel of Figure 1—normalized by the total number of articles, denoted as the set $\mathcal{A}$. A higher index value indicates more intense shortages. Below, we outline the steps used to create the search query, reported in Table 1, for isolating set $\mathcal{S}$.

First, we construct a broader set of articles, $\mathcal{B}$, the pink area in Figure 1. Articles in this set must mention at least one shortage term—namely 'shortage,' 'scarcity,' 'bottleneck,' or 'rationing'—in conjunction with an economics-related term, such as 'economy,' 'market,' or 'commerce.' The shortage terms used are those most frequently linked to economically relevant constraints on production or on the availability of goods to consumers.[2] Including an economics-related term reduces the occurrence of false positives—articles mentioning shortages unrelated to economic phenomena, such as shortages of empathy or goodwill.

Second, we draw a random sample of about 3,300 articles from the set $\mathcal{B}$ and extract the 1,000 most frequent collocates, words appearing within five words of the shortage terms. From these collocates, we select those indicative of shortages in specific sectors. The most common collocates, excluding stopwords, include 'oil,' 'water,' 'war,' 'time,' 'coal,' 'days,' 'food,' 'cars,' 'people,' 'government,' 'million,' 'labor,' 'state,' 'home,' 'steel,' and 'fuel.' Subsequently, generic

[2] Potential synonyms of shortage, such as 'lack,' 'paucity,' or 'insufficiency,' were excluded because they have a broader range of meanings and are less likely to be specifically associated with economic shortages.

words like 'time,' 'days,' 'people,' and 'government,' which convey little information about shortages, are removed. The remaining collocates are grouped into four topics: food, industry, labor, and energy.[3] This classification is supported by a Latent Dirichlet Allocation (LDA) analysis (Blei et al., 2003), an unsupervised machine learning technique that identifies latent topics by analyzing word co-occurrences. We apply LDA to a random sample of nearly 14,000 newspaper abstracts mentioning shortages. The analysis confirms our four-topic taxonomy, with identified topics corresponding to energy, food and agriculture, industrial goods, and labor markets (see bottom panel of Figure 1). Details on the LDA methodology, preprocessing steps, and topic distributions are provided in the Appendix.

Using the lists of shortage terms, economic terms and topic-specific terms, we construct the search query shown in Table 1. An article is included in the shortage index (set $\mathcal{S}$ in Figure 1) if it meets two conditions: (1) A shortage term must appear within five words of a topic-specific term, and (2) the article must contain at least one economics term. If an article meets the first condition for multiple topic-specific categories, it is counted once for each—so the total number of shortage articles is the sum across categories. This method gives greater weight to articles discussing multiple types of shortages, enhancing the index's informational content. Section 3 documents that the requirement of proximity between shortage and topic-specific terms is crucial to reducing false positives and improving search accuracy.

2.2 Shortages in History

Figure 2 shows the monthly shortage index from 1900 through 2024. The index is calculated as the monthly share of articles discussing economic shortages (number of articles in $\mathcal{S}$ divided by the number of articles in $\mathcal{A}$), scaled so that its average is 100 between 1900 and 2019. Figure 3 presents a breakdown of the index by topic. Table 2 lists the 30 largest spikes in the index, along with descriptions of the key events associated with each episode.[4]

The index varies considerably over time, with the most notable spikes linked to events related to the four classification topics—energy, food, industry, and labor. These episodes

[3] Exploratory work identified additional categories (housing, health, and education shortages), but these were excluded because they either reflect long-standing social issues or contribute minimally to the evolution of the overall index.

[4] We calculate spikes by first extracting the residuals from a regression of the shortage index, h_t, on its values two months before and two months after. The largest residuals are our candidate spikes. We then discard any spike if it is not a local maximum within the 13-month window $[t-6, t+6]$, ensuring that we capture true local peaks. The table reports the 30 largest residuals.

typically reflect a combination of traditional supply disruptions, demand pressures, and market-clearing frictions, such as price controls, logistics breakdowns, or coordination failures, that prevent normal price adjustment mechanisms from equilibrating supply and demand. Adverse geopolitical events, especially wars, are strongly associated with severe shortages and often feature multiple such frictions simultaneously. For instance, the index rises sharply during World War I and during World War II, peaking at over 1,000 (10 times the sample mean) in January 1943, with spikes recorded across all shortage categories. Other events, such as the Suez Crisis and the Iraqi invasion of Kuwait in 1990, also coincide with substantial spikes.[5] Many of these spikes are linked to energy shortages, often resulting from wars and instability in the Middle East, such as during the oil shocks of the 1970s. Labor shortages, particularly those caused by strikes, were historically significant, especially in the early part of the sample. For example, coal-related strikes in 1903, 1919, and 1922 produced notable spikes. The labor component of our shortage index exhibits a strong relationship with independent measures of labor market tightness. Michaillat and Saez (2024) identify four episodes when the U.S. labor market was inefficiently tight or beyond full employment: World War II, the Korean War, the Vietnam War, and the post-pandemic period. These episodes nicely align with spikes in the labor component of our index.

More recently, the index spiked on several occasions the COVID-19 pandemic. The first spike corresponds to shortages in medical equipment and healthcare workers at the pandemic's onset. A second, larger spike occurred at the beginning of 2022, driven by global supply bottlenecks as economies reopened after prolonged mobility restrictions in 2020 and 2021. As shown in Figure 3, this second spike primarily reflected shortages of labor and industry components.

The stacked bar chart at the bottom of Figure 3 visualizes the probabilistic topic mixture calculated with the LDA algorithm for each abstract, sorted by year. Each year's news coverage reflects varying degrees of focus on different shortage topics, confirming the taxonomy of shortages from the analysis of the sub-indexes. In the early part of the sample, discussions are primarily centered around industry-related shortages, with food and water shortages receiving some attention. Energy shortages become prominent during the 1970s during oil crisis episodes. In the more recent period, following the COVID-19 pandemic, labor shortages have increasingly dominated the news, reflecting the widespread disruptions in workforce availability and supply

[5] Federle et al. (2024) show that wars are associated with supply disruptions both domestically and abroad.

chains that persisted post-pandemic.

2.3 Key Drivers of Shortages and Breakdown by Country

To determine whether shortages tend to occur alongside particular phenomena, we examine their co-occurrence in the news with other economic events—specifically, labor strikes, natural disasters, adverse geopolitical events, pandemics, and price controls. Figure 4 compares the actual joint probability of these events being mentioned with shortages to the probability expected under independence. The lift ratio, defined as the ratio of the observed joint probability to the product of the individual probabilities, serves as a measure of association. A lift ratio greater than one indicates that these events co-occur more frequently than would be expected by chance.

Price controls have the strongest association with shortages, co-occurring 7.6 times more frequently than expected under independence. Other disruptive events, such as strikes, natural disasters, geopolitical conflicts, and pandemics, also show meaningful associations, co-occurring between two and three times more often than expected. These events often involve not only traditional supply disruptions or demand shifts, but also market-clearing frictions such as logistics breakdowns, distribution bottlenecks, and coordination failures that prevent available goods from reaching buyers efficiently.

Figure 5 breaks down the shortage index by country. Specifically, we "slice" the main index by counting the number of articles that satisfy the shortage search query and mention either the name of a country or one of its major cities. As expected, the country-specific shortage indexes all tend to spike around global events. However, countries are differentially exposed to global shortages in terms of timing—for instance European countries experienced earlier and more prolonged shortages during the two world wars than North American countries—and intensity, as evidenced by France and Germany during World War I and by Japan in the late 1970s.

The figure also reveals some country-specific episodes of shortages. It shows oil and energy shortages in France and the United Kingdom following the Suez Crisis, concerns about shortages in Germany around the time of the construction of the Berlin Wall in 1961, or shortages in the aftermath of the 2011 tsunami in Japan.[6] In sum, the country-specific indexes offer important

[6] In line with our measurement, the Suez Crisis did not create major energy shortages in Germany because its energy supply was less reliant on Middle Eastern sources compared to France and the United Kingdom.

insights for comparative and cross-country research. Their ability to capture country variation in shortage events reinforces the robustness of our search-query-based approach.

3 Assessing the Accuracy of the Shortage Index

We perform two exercises to assess the index's accuracy. First, we confirm that the newspaper articles in the index mention actual shortage-related concerns. Second, we evaluate whether the index aligns with alternative proxies for shortages during the overlapping periods. A key advantage of our index is its long historical coverage—from 1900 through present—compared to existing measures that span only shorter, more recent time frames.

3.1 Validation of the Shortage Index

Audit of the Overall Index

We verify that our index accurately measures shortages—minimizing both Type-I and Type-II errors—by conducting a two-part text audit. To evaluate Type-I errors (false positives), we sample the abstracts of 872 articles from the shortage set $\mathcal{S}$.[7] To evaluate Type-II errors (false negatives), we examine a separate sample of 298 articles outside of $\mathcal{S}$, as detailed later in this section. By design, each article in $\mathcal{S}$ contains at least one economics-related term along with a reference to shortages, scarcities, or bottlenecks near a topic word (energy, food, industry, or labor). For each article, we extract the first snippet of text that references shortages. These snippets are centered on the shortage word and are limited to 110 characters—a length chosen to balance brevity as well as computational and cognitive load. For example, one sample snippet is:

> "...Ford motor company said today that still lingering steel shortages would force a closing next week of eighteen..." (New York Times, 1952)

We then used Claude 3.7 Sonnet reasoning model (Anthropic, 2024) to determine whether each snippet mentions current or prospective shortages, rationing, scarcity, or bottlenecks

[7] The abstracts must also satisfy the conditions to belong in the set $\mathcal{S}$. Our search query is run on the six U.S. newspapers listed in Section 2. Abstracts for the six newspapers are only available until 2014. For the period 2015-2024, we sampled abstracts from a broader set of newspapers, including British, Canadian, and Australian newspapers. Abstracts in the database are typically short portions of text, often containing the opening sentences or the initial paragraphs of the article. A list of the abstracts is available at https://www.matteoiacoviello.com/shortages_files/claude_audit_report.html.

related to industry, labor, materials, food, or water. Claude returned a table of results, coding snippets as 1 (shortage mentioned), 0 (shortage not mentioned), or 99 (unsure). Additionally, Claude gave a brief explanation for each coding decision.

Before initiating classification, we provided Claude with examples—listed in the Appendix—of how to code the snippets, ensuring that the training sample included false positives (e.g., mentions of the lack or end of shortages). Although AI-based validation is not foolproof, we found that Claude performed comparably to a human evaluator. To ensure the reliability of the AI classification, we manually reviewed a random subsample of the snippets and found overwhelming agreement between human coding and Claude's output. Claude even demonstrated the ability to extrapolate contextual cues (such as linking a sentence to a particular country or individual). For example, when processing the snippet "Economy may be slowing but Lowe is banking on labour shortages gradually leading to an increase...," Claude classified it as 1, and noted that the Reserve Bank of Australia was expecting labor shortages to lead to wage growth.[8]

The audit results are summarized in Table 3. Out of 872 articles in the set $\mathcal{S}$, only 6.3 percent were classified by Claude as false positives. For example, Claude coded the snippet "...says that while there is not necessarily a shortage of people wanting to work in management" as 0, explaining that there was "No shortage of people wanting to work in management." We then repeated the audit for a sample of 298 articles outside the set $\mathcal{S}$. Among these, only one article mentioned shortage of manufacturing plants that were not captured by our search query ("recycling of newsprint was held back by a shortage of de-inking plants"). Notably, our search query deliberately excluded the word "plants" because preliminary tests revealed that its inclusion produced numerous false positives.[9]

As a final check, we confirmed that restricting the search to include shortage words near key terms related to industry, labor, food, or energy significantly improves accuracy. In a broader set of articles in which the presence of shortage words near these key terms is allowed but not required, 84.2 percent of articles mention economically relevant shortages. This approach yields a Type I error rate of 15.8 percent, considerably higher than that achieved with our preferred search query. False positives in the broader set include articles referencing non-economic

[8] The excerpt quoted is from March 2019, when Mr. Philip Lowe was Governor of the Reserve Bank of Australia.

[9] For example, see the article "Brighten Up Indoors with Colorful Plants" (Los Angeles Times, February 4, 1996) that states "There's no shortage of plants with brightly colored foliage to liven up your kitchen."

shortages, such as shortages of political campaign funds, a lack of quality baseball photos, legislative bottlenecks, and shortages of sunshine.

Empirical Validation of Categorical Shortage Indexes

Although the categorical shortage sub-indexes are correlated, each captures distinct, sector-specific variation. We regress selected measures of sectoral price pressures on the four components of the shortage index. Labor shortages predict earnings growth, the industry shortage index predicts producer price index (PPI) inflation, and the energy and food indexes predict their respective consumer price index (CPI) inflation components. Because each sectoral index correlates precisely with the prices and wages most exposed to that sector's shortages, the results confirm that the disaggregated components contain valuable informational content beyond the aggregate measure. The Appendix reports the full table of these regression estimates. Collectively, these empirical results and the text-based audits validate that the overall shortage index is a robust, economically important measure of scarcity rather than a statistical artifact.

3.2 Comparison with Other Indicators of Supply Constraints

Our shortage index differs from existing supply-constraint measures in two key dimensions: sectoral scope and historical coverage. Figure 6 compares our index with three prominent alternatives: the New York Fed Global Supply Chain Pressure Index (GSCPI) (Benigno et al., 2022), the ISM Supplier Deliveries Index (SDI), and the Supply Bottlenecks Index (SBI) for the United States (Burriel et al., 2023). The GSCPI focuses on transportation and shipping costs through freight rates and delivery times. The SDI measures supplier delivery delays from purchasing manager surveys. The SBI uses newspaper searches targeting articles mentioning "supply" or "supply chain" alongside disruption terms. Our index captures shortage episodes across four sectors—energy, food, industrial goods, and labor—without requiring supply chain framing, enabling analysis of how different shortage types affect macroeconomic outcomes. We standardize each variable to have mean zero and unit variance over the overlapping period.

Our index co-moves strongly with these measures during major disruptions. The correlation with the GSCPI (1997–present) is 0.73, with both spiking during COVID-19. The correlation with the SDI is 0.26, though both capture the 1979 oil crisis and post-pandemic disruptions. Our index correlates at 0.90 with the SBI (1990–present), reflecting strong alignment during

their overlapping period, which is dominated by COVID-19—a supply chain disruption affecting multiple sectors simultaneously. However, our sectoral breadth allows us to capture episodes not framed in supply chain terms, such as early 20th century labor strikes, wartime price controls and rationing, or geopolitical energy disruptions.

The 1990 Iraq invasion of Kuwait illustrates these distinctions. Our index and the SBI spike modestly and briefly, while the ISM index declines sharply, primarily reflecting the concurrent recession. The modest, transitory spike accurately reflects that while energy supply faced a binding constraint prompting Strategic Petroleum Reserve releases and increased Saudi production, policy interventions prevented prolonged shortages.

A second advantage of our index is historical coverage. The GSCPI begins in 1997, the SBI in 1990, and the SDI in 1948. Our index extends to 1900, providing 125 years across major episodes—World War I, World War II, the Great Depression, and events that shaped economic policy and institutions. This is particularly valuable given that the post-1990 period (excluding COVID-19) experienced few severe shortage episodes, limiting variation in more recent measures. Our subsequent analysis shows that the shortage-inflation relationships vary across regimes invisible to post-1990 indices.

4 Shortages as Predictors of Inflation and Activity

In this section, we analyze how shortages relate to inflation and economic activity using a predictive regression framework. Our results indicate that higher levels of shortages are associated with prolonged periods of elevated inflation and lower economic activity. The effects vary in size over time, underscoring the macroeconomic significance of quantity constraints across more than a century of U.S. data.

4.1 Predictive Regressions: Shortages, Inflation, and Activity

We explore the relationship among shortages, inflation, and economic activity using the following predictive regression:

$$\Delta Y_{t+h} = \alpha + \beta\, SHORTAGE_t + \sum_{i=0}^{p} \boldsymbol{\gamma}_i' \mathbf{X}_{t-i} + \varepsilon_{t+h} \tag{1}$$

where

$$\Delta Y_{t+h} = \frac{400}{h} \ln \left(\frac{Y_{t+h}}{Y_t} \right)$$

represents the annualized log change of a variable of interest Y_t between period t and horizon h, and $SHORTAGE_t$ denotes the shortage index. The vector $\mathbf{X}$ contains control variables. For this exercise, we use quarterly BEA data from 1950 through 2024 for real per-capita GDP, personal consumption expenditures, and private fixed investment. Inflation for each category is measured using the associated price deflator.

For each price and economic activity indicator, we estimate equation (1) by OLS using heteroskedasticity-robust standard errors as recommended by Montiel Olea et al. (2025).[10] As control variables, we include two lags of the shortage index and quarterly changes of the dependent variable and the corresponding economic indicator or price deflator, both contemporaneously and with two lags. For example, the regression for real GDP growth includes contemporaneous and lagged values of both real GDP growth and inflation (measured by the log change in the GDP deflator).

Figure 7 presents the results for four-quarter-ahead regressions (the Appendix reports results for the one- and eight-quarter horizons). For ease of comparison, we report standardized estimates of the coefficient β. In the full sample, higher shortages are associated with a rise in inflation across GDP components. The effects on economic activity show substantial heterogeneity: consumption of both durable and nondurable goods and private investment decline significantly, while services consumption and aggregate GDP show small and statistically insignificant effects. Quantitatively, the standardized coefficients imply that, for instance, a one-standard-deviation increase in the shortage index is associated with a rise in durable goods inflation of 0.6 percentage points and a decline in durable goods growth of 2 percentage points.

Figure 7 also documents substantial time variation in these relationships. When we split the sample into two periods, pre- and post-2000, shortages exhibit a positive correlation with future sectoral inflation throughout, but their link with economic activity shifts from significantly negative in the earlier period to significantly positive in the latter across all components. Thus, the weak and insignificant full-sample coefficients on services and aggregate GDP reflect averaging across these two periods with offsetting signs rather than a true absence of relationship. This shift reflects changes in the underlying drivers of shortages across periods.

[10] Results are qualitatively similar using heteroskedasticity and autocorrelation consistent standard errors following Newey and West (1987).

As we show in Section 5, shortages respond endogenously to both demand and supply shocks: they rise when supply is disrupted or when demand surges faster than supply can adjust. The positive correlation between shortages and activity during the more recent period, which includes the COVID-19 pandemic, suggests that demand-driven shortages—where strong aggregate demand meets capacity constraints—became relatively more prominent compared to the pre-2000 period, when supply disruptions and market-friction shocks played a larger role.

A further concern is whether the shortage index contains information beyond that captured by other macroeconomic variables. We address this issue by augmenting our baseline regression with additional controls, including oil prices, commodity prices, wage growth, and inflation expectations. The inclusion of these controls only slightly attenuates the effect of shortages on inflation and activity, but the coefficients remain in large part statistically significant (see the Appendix for details). These findings suggest that our shortage index contains additional information beyond that captured by traditional macroeconomic indicators.

4.2 Time Variation in Shortage Effects: Rolling Regressions

To assess the stability of our findings across different historical periods, we estimate equation (1) using rolling 30-year backward-looking windows. Each point on the chart at date t shows the coefficient estimated using data from the preceding 30 years. For example, the coefficient shown for 1973 is based on data from 1943 through 1973. We focus on real GDP growth and its deflator, extending the sample back to 1900:Q1.[11]

Figure 8 reveals stability in the shortage-inflation relationship alongside substantial variation in the shortage-activity relationship. Shortages consistently predict higher inflation starting in the 1940s, with coefficients remaining positive and statistically significant across nearly all rolling windows. The effect becomes negligible only during the decade preceding the pandemic, when shortage episodes were rare. This positive relationship holds across diverse historical episodes—including both world wars, the 1970s oil shocks, and the post-pandemic period.

The relationship with economic activity, by contrast, exhibits marked time variation. Coefficients are positive but generally insignificant in rolling windows through the 1960s, with the exception of periods when World War II—an episode with large positive and significant effects—is included in the sample. Once World War II exits the 30-year window and estimates

[11] We extend the sample for GDP, the GDP deflator, and population using data from Ramey and Zubairy (2018).

are based primarily on post-WWII data, coefficients turn significantly negative, remaining so from the early 1970s through the early 2000s. In the most recent period, coefficients become positive and significant again. This time-varying pattern suggests that the dominant drivers of shortage episodes differ across historical periods: the post-pandemic era features episodes where strong demand pressures alongside binding supply constraints elevated both shortages and output, while the intermediate period saw shortages arising primarily from supply disruptions and market frictions that constrained activity.

The results in this section show that shortages contain predictive information for both inflation and economic activity, with effects that vary across historical episodes. To identify the underlying forces driving these comovements, the next section develops a structural framework that decomposes fluctuations in shortages and macroeconomic outcomes into economically interpretable shocks.

5 Shortages: A VAR Analysis

The predictive regressions presented in the previous section establish a correlation between our shortage index and subsequent macroeconomic realizations. However, these estimates cannot disentangle the latent drivers of shortages, whether they stem from exuberant demand, impaired productive capacity, or disruptions to the market-clearing mechanism itself. To address these issues, we estimate a structural Vector Autoregression (SVAR) model that decomposes the joint dynamics of shortages, activity, and inflation into their underlying components. Our approach follows the Bayesian framework of Baumeister and Hamilton (2019). This methodology is particularly suited to our context as it avoids the "all-or-nothing" nature of point-identified exclusion restrictions, instead providing a posterior distribution of structural parameters that reflects both the empirical data and the bounds of economic theory. By explicitly modeling the shortage wedge, we provide a framework for understanding how supply and demand shocks propagate through a system where prices may not adjust instantaneously to clear all markets.[12]

[12] An alternative approach would involve estimating a VAR with nonlinearities or time-varying parameters. However, given the lack of a clear consensus regarding the exact form of such nonlinearities, we view the standard structural VAR as a necessary conceptual prerequisite.

5.1 The Model

The VAR model includes three quarterly variables: inflation, activity, and shortages. These are summarized by the vector:

$$\mathbf{X}_t = (\pi_t, y_t, h_t)' \tag{2}$$

where inflation (π_t) and activity (y_t) are quarter-over-quarter percent changes in the U.S. headline CPI index and real GDP, respectively; and shortages (h_t) are expressed in levels using our standardized shortage index.[13] All series are expressed in deviation from their sample mean. The sample runs from 1950:Q1 through 2024:Q4.

The system of equations consists of a Phillips Curve, an aggregate demand equation, and an equation connecting shortages with aggregate demand and aggregate supply. The structural representation is given by:

$$\mathbf{A}_0\mathbf{X}_t = \sum_{j=1}^{p} \mathbf{B}_j\mathbf{X}_{t-j} + \mathbf{u}_t, \tag{3}$$

where $\mathbf{u}_t = (u_t^S, u_t^D, u_t^H)'$ is the vector of structural shocks—supply, demand, and shortage shocks—with zero mean and a diagonal covariance matrix $E[\mathbf{u}_t\mathbf{u}_t'] = \mathbf{\Sigma}_u$.

The structural relationships among variables, with $\mathbf{z}_{t-1} = (\mathbf{X}_{t-1}', \mathbf{X}_{t-2}', \ldots, \mathbf{X}_{t-p}')'$ denoting the vector of lagged endogenous variables, are described by the following equations:

$$\pi_t = \kappa\ y_t + [\mathbf{b}^S]'\mathbf{z}_{t-1} + u_t^S, \tag{4}$$

$$y_t + \alpha\ h_t = -\delta\ \pi_t + [\mathbf{b}^D]'\mathbf{z}_{t-1} + u_t^D, \tag{5}$$

$$h_t = \phi_S\ u_t^S + \phi_D\ u_t^D + [\mathbf{b}^H]'\mathbf{z}_{t-1} + u_t^H. \tag{6}$$

Equation (4) represents the economy-wide aggregate supply relationship, where inflation is positively related ($\kappa > 0$) to measured activity y_t and subject to supply shocks u_t^S.

Equation (5) characterizes the aggregate demand relationship, augmented to incorporate a shortage wedge. As in Baumeister and Hamilton (2018), desired demand d_t is assumed to

[13] We deliberately maintain a parsimonious three-variable specification to transparently isolate the shortage wedge and avoid the curse of dimensionality inherent in larger models. In an earlier version of this draft (Caldara, Iacoviello, and Yu, 2025), we estimated a higher-dimensional systems including the federal funds rate and commodity prices. That specification shows that shortage shocks are different from standard monetary policy or commodity disturbances, which manifest as traditional shifts in aggregate demand or supply in our three-equation model.

depend negatively on inflation with an elasticity of δ. We assume that shortages create a gap between desired demand and realized activity y_t, so that desired demand can be expressed as $d_t \equiv y_t + \alpha h_t$, where $\alpha > 0$ is a scaling parameter converting the shortage index into units of economic activity. When shortages are elevated ($h_t > 0$), desired demand d_t exceeds realized activity for a given inflation rate π; by contrast, when shortages are low ($h_t < 0$), supply exceeds demand. The shortage wedge αh_t thus quantifies unmet demand—the difference between what economic agents wish to purchase and what actually gets produced and sold. This gap arises when frictions prevent prices from adjusting sufficiently to equate supply and demand. Such frictions may include institutional rigidities (e.g., price controls), search and matching inefficiencies in product or input markets, or any temporary capacity constraints that cannot be quickly resolved. The specification captures an economy where market-clearing occurs through a combination of price adjustment in some markets and quantity rationing in others.

Equation (6) models the shortage index. Traditional approaches to modeling demand and supply in multivariate models implicitly assume that markets clear instantly, so that $h = 0$ at all times and production always coincides with actual demand. Here we relax this assumption by modeling shortages as a direct function of the structural shocks hitting the economy. In particular, shortages increase when demand surges ($\theta_D > 0$) or when supply is disrupted ($\theta_S > 0$). Implicit in this formulation is the assumption that a positive shock to demand can spill over into shortages rather than being absorbed by price or quantity adjustments. Similarly, supply shocks may result in shortages to the extent that prices do not fully rise to absorb the decline in production.

The shortage equation includes an exogenous structural shock u_t^H. This shock u_t^H captures changes in the economy's capacity to facilitate exchange—shifts in the ability to connect demand with available supply that are not fully captured by standard supply and demand disturbances. Unlike traditional demand and supply shocks that represent changes in preferences, technology, or productive capacity, u_t^H reflects disturbances to the market mechanism itself. Examples include sudden regulatory interventions (such as emergency price controls), logistical disruptions that segment markets geographically or temporally, coordination failures (such as panic buying or hoarding behavior), or rapid deterioration in the effectiveness of distribution networks. Major economic events—such as wars, natural disasters, or large-scale strikes—typically involve a combination of all three shocks in our framework. For instance, wartime episodes often

feature genuine reductions in productive capacity (supply shocks u_t^S) alongside price controls and distribution network disruptions (shortage shocks u_t^H). Similarly, the 2011 Japan tsunami disrupted productive capacity but also created logistical bottlenecks and coordination problems that prevented available supply from reaching buyers efficiently (both u_t^S and u_t^H).

5.2 Identification and Estimation

Following Baumeister and Hamilton (2019), we treat identification and estimation of this structural VAR model as a Bayesian exercise, imposing prior restrictions directly on the structural parameters $(\kappa, \alpha, \delta, \theta_S, \theta_D)$ and the lag coefficients in $[\mathbf{b}^S]'$, $[\mathbf{b}^D]'$, and $[\mathbf{b}^H]'$. Priors on the contemporaneous structural parameters discipline the identification of structural shocks based on economically motivated sign restrictions, while priors on lagged coefficients serve to regularize the dynamics of the system. We estimate the model with $p = 2$ lags. This choice balances parsimony with flexibility; estimating a more flexible specification with 8 lags has negligible effects on our main results, as shown later in Figure 13. Additional details on prior distributions for all model parameters, along with their posterior means, standard deviations, and 80 percent credible intervals are in the Appendix.

Regarding the supply and demand slopes, the priors for κ and δ in equations (4) and (5) follow inverse gamma distributions with a mean of 0.2 and a standard deviation of 1. At these values, a 1 percent demand-driven increase in activity raises inflation by 0.2 percentage points, and a 1 percent supply-driven increase in inflation reduces GDP growth by 0.2 percentage points. These assumptions are broadly consistent with estimates from both DSGE models (Smets and Wouters, 2007) and structural VAR models of the U.S. economy (Baumeister and Hamilton, 2018). Similarly, the shortage response parameters θ_S and θ_D share inverse gamma priors with a mean of 0.2 and a standard deviation of 1. These priors impose that, consistent with standard economic reasoning, shortages increase in response to contractionary supply shocks and expansionary demand shocks. For the shortage scaling parameter α, we assign a beta prior with mean 0.25 and standard deviation 0.125, a value that implies modest but meaningful shortage effects on the activity gap, while allowing substantial updating by the data.

We set the priors for the lagged structural coefficients to ensure the model dynamics are empirically grounded while allowing the data to provide substantial updating. Specifically, we estimate an auxiliary VAR(2) on the transformed variables $\mathbf{A}_0(\bar{\theta})\mathbf{X}_t$, where the contemporane-

ous matrix $\mathbf{A}_0$ is evaluated at the prior means of the structural parameters $\{\delta, \kappa, \alpha, \theta_S, \theta_D\}$. We set the prior means for the diagonal elements of the structural lag matrices $\mathbf{B}_j$ to the OLS point estimates derived from this regression. This procedure centers the prior in a region of the parameter space consistent with the observed persistence of the data, while avoiding draws that imply unstable or nonstationary dynamics. The use of OLS-based dispersion ensures that the prior remains diffuse, so that the likelihood of the data is the primary driver of posterior inference.[14]

The off-diagonal (cross-lag) coefficients are assigned prior means of zero with a standard deviation of 0.05, shrinking the model toward own-lag dynamics, in line with standard Minnesota-type priors. The sole exception is the coefficient on lagged shortages in the inflation equation, $\rho_{1,\pi H}$, which we set to 0.10. At the prior mean, this value ensures that a shortage shock leaves inflation unchanged in the first year on average, while letting the data speak regarding the overall inflationary effect of shortages. As is standard in Bayesian VAR analysis, these priors serve primarily to regularize the dynamic coefficients and restrict attention to stationary regions of the parameter space, rather than to impose economically meaningful cross-equation restrictions.

In a similar vein, we set the prior means for the standard deviations of the structural shocks, σ_i, to the estimated standard deviations of the residuals from the auxiliary VAR. We assume an Inverse-Gamma distribution for these parameters, $\sigma_i \sim IG\left(\hat{\sigma}_i^{OLS}, \frac{1}{2}\hat{\sigma}_i^{OLS}\right)$, where $\hat{\sigma}_i^{OLS}$ is the OLS estimate. This specification provides a centered but relatively diffuse prior for the volatility of the structural innovations.

Priors and posteriors for key structural parameters are shown in Figure 9. All prior densities are intentionally wide to allow the data to update both the location and spread of these distributions.

5.3 Results

We estimate the model using the algorithms in Dynare (Adjemian et al., 2011). The first 20 observations serve as a training sample, and we draw 50,000 posterior samples using a random

[14] To formalize our priors on the dynamic coefficients, let $\rho_{j,km}$ denote the structural coefficient on the j-th lag of variable m in the equation for variable k, where the subscripts $k, m \in \{\pi, Y, H\}$ correspond to the variables π_t, y_t, and h_t. These scalars constitute the elements of the lag vectors $\mathbf{b}^S$, $\mathbf{b}^D$, and $\mathbf{b}^H$. For example, the lagged terms in the aggregate supply equation can be explicitly written as $[\mathbf{b}^S]'\mathbf{z}_{t-1} = \sum_{j=1}^{p}(\rho_{j,\pi\pi}\pi_{t-j} + \rho_{j,\pi Y}y_{t-j} + \rho_{j,\pi H}h_{t-j})$.

walk Metropolis-Hastings algorithm.

Parameter Estimates

Figure 9 shows the prior distributions and the estimated posterior distributions for the key parameters of the model.[15] The location and spread of all posterior distributions are substantially updated compared to the priors, revealing that the data and the structure of the model are informative about the structural parameters.

The posterior mean for the shortage wedge parameter α is 0.19. This value indicates that when the shortage index is one standard deviation above its mean, production falls short of demand by approximately 0.19 percentage points. This effect underscores the macroeconomic significance of the shortage wedge in constraining realized output below desired demand. The posterior distribution is substantially updated from the prior, indicating that the restrictions and the data provide information about how shortages translate into output.

The posterior mean for κ is around 0.12—a demand shock that raises GDP growth by 1 percentage point leads on impact to a rise in inflation of around 0.12 percentage points. The posterior mean for δ in the aggregate demand equation is 0.14, which implies that a supply shock that increases inflation by 1 percentage point leads to an impact decrease in GDP growth of about 0.14 percentage points.[16] All told, the posterior distributions reveal a "flatter" supply curve—lower sensitivity of inflation to shifts in demand—and a "steeper" demand curve—higher sensitivity of inflation to shifts in supply—than assumed by the prior distributions.

Estimates for θ_D (around 0.06) and θ_S (around 0.19) are consistent with the notion that shortages arise in response to demand and supply shocks, more so for supply than for demand shocks. While a full interpretation of the relative magnitudes is difficult absent a full structural model, one way to read this result is that prices adjust to clear the market more slowly in response to demand shocks than in response to supply shocks.

The bottom right panel of Figure 9 illustrates one of the underlying forces driving the inflationary effects of shortages in our model. After controlling for current and lagged movements in all other variables, the posterior density measuring the effect of shortages in period t on

[15] Table A.2 shows the prior distributions for all model parameters, along with their posterior mean, standard deviation, and 80 percent credible intervals.

[16] The supply shock also affects shortages contemporaneously through (6). This feedback effect, while quantitatively small, reinforces the GDP decline.

inflation in period $t+1$—as captured by the parameter $\rho_{1,\pi H}$—shows that most of its probability mass lies in the positive region. This finding indicates a high likelihood that lagged shortages have a positive effect on inflation.

Impulse Responses

Figure 10 reports the impulse responses to the estimated shocks in the model.

An exogenous shortage shock leads to a persistent rise in inflation alongside a slow, persistent decline in GDP. Quantitatively, a one-standard-deviation shortage shock leads to a rise in inflation after one year of about 0.2 percentage points. As shortages return toward baseline, inflation continues rising and peaks about two years after the shock, remaining elevated thereafter. Shortages also respond endogenously to other shocks: an expansion in aggregate demand or a contraction in aggregate supply lead to higher shortages. All told, we find bidirectional causality: exogenous shortage shocks trigger business cycle fluctuations, and traditional macroeconomic shocks generate movements in shortages.

We highlight two insights from our impulse response analysis. First, a shock that leads to an immediate jump in shortages has significant and persistent effects on GDP and inflation, with the maximum effects occurring between one and two years after the shock. The size and timing of these effects highlight the value of the shortage index as a leading indicator for macroeconomic variables. Second, compared to a traditional contractionary supply shock, a shortage shock produces distinctly different inflation dynamics relative to its impact on economic activity. For a one-percentage-point rise in inflation, GDP growth bottoms out at about 1 percent after a shortage shock, compared to a 0.33 percent decline after a supply shock. This pattern reflects the defining characteristic of shortage shocks in our framework: they initially act as demand wedges that constrain activity, generating larger output effects relative to their inflationary impact. The persistent rise in inflation due to shortage shocks results from the feedback channel captured by $\rho_{1,\pi H}$, as shortages can lead to scarcity, pricing disruptions, and increased operating costs.

Finally, Figure 10 shows that the prior-induced impulse responses have much wider confidence intervals than the corresponding posterior intervals, indicating that the data are informative about the model parameters.

The Effects of Shortages throughout History

Figure 11 presents the historical decomposition of the model, illustrating how the three structural shocks have contributed to movements in shortages, GDP growth, and inflation over time. The decomposition reveals that the shortage index is primarily driven by shortage shocks themselves. However, traditional demand and supply shocks contribute meaningfully to shortage dynamics around key historical episodes. For instance, in the early 1950s, shortages reflect a combination of demand shocks associated with the rapid transition from wartime to peacetime economies and shortage-specific disruptions related to the Korean War. The 1970s oil embargo period shows elevated shortage shocks that contributed to both higher inflation and slower economic activity, alongside the traditional supply shocks associated with energy price increases. Shortages remained muted throughout the 1960s and most of the 1990s and early 2000s, before spiking dramatically during the COVID-19 pandemic and its aftermath.

Importantly, the historical decomposition supports our interpretation of shortage shocks as capturing distinct disturbances beyond traditional supply and demand forces. Even during periods when shortage shocks matter for macroeconomic outcomes—such as the 1970s and the post-pandemic period—they typically operate alongside rather than instead of traditional shocks, and are not the dominant driver of economic fluctuations.

To better highlight the macroeconomic impact of shortages in recent years, Figure 12 zooms in on the pandemic and post-pandemic period. The period 2020:Q1–2021:Q1 features adverse shortage shocks driven by labor market disruptions and supply chain bottlenecks, alongside depressed aggregate demand. Shortages remain low because adverse shortage shocks are offset by weak demand, leaving inflation close to baseline. The period 2021:Q2–2022:Q3 shows all three shocks contributing to macroeconomic outcomes: persistent shortage shocks continue to depress activity, traditional supply shocks exert upward pressure on inflation with modest output effects, and strengthening demand shocks adds inflationary pressures. The period 2022:Q4–2024:Q4 is marked by declining shortages, though they remain above pre-pandemic levels. Importantly, given the delayed and persistent effects of shortage shocks, past shortages continue contributing to elevated inflation well into 2024.

This decomposition reinforces our interpretation of shortage shocks as capturing disturbances distinct from traditional supply and demand forces. During the pandemic period, shortage shocks reflect well-documented disruptions—the goods-versus-services demand rebal-

ancing, port congestion, semiconductor bottlenecks, and shipping constraints—that created mismatches between the composition of demand and available supply beyond what aggregate demand or supply shifts alone could explain. All told, the exercise illustrates that shortage shocks account for a non-negligible share of the post-pandemic rise in U.S. inflation.

Most of the discussion in policy and research about the drivers of inflation and economic activity since 2020 has been framed in terms of demand and supply forces. To connect our findings to this debate, Figure 13 presents historical decompositions under four model specifications: our baseline estimated model (top row), a version using parameters calibrated at the prior means (second row), a restricted specification with a steep supply curve ($\kappa = 1$, third row), and a more flexible specification with 8 lags instead of 2 (bottom row). The bottom panel demonstrates that the results are robust to increasing the number of lags: the decomposition with 8 lags is nearly identical to the baseline, confirming that a parsimonious 2-lag specification captures the essential dynamics. This comparison reveals two key insights about the role of shortage shocks in post-pandemic inflation.

First, the role of all shocks in explaining inflation dynamics depends critically on the estimated slopes of the supply and demand curves. In the restricted specification with a steep supply curve ($\kappa = 1$), the model simultaneously estimates a flatter demand curve. Under this configuration, demand shocks become the dominant driver of post-pandemic inflation, consistent with the findings of Giannone and Primiceri (2024) for the euro area and the United States. Our baseline estimates—featuring a flatter supply curve and steeper demand curve—attribute a larger share of inflation dynamics to supply-side disturbances, including both traditional supply shocks and the disruptions captured by shortage shocks. The difference in shock decompositions across specifications reflects an important tension in interpreting recent inflation: models that emphasize demand pressures feature relatively flat demand curves, while models that emphasize supply constraints and matching frictions require steeper demand responses to price changes. The data favor our baseline specification, as indicated by its higher marginal likelihood.

Second, the contribution of shortage shocks to inflation is strongly informed by the data rather than imposed by our priors. At the prior mean, shortage shocks account for none of the post-pandemic inflation surge; the model attributes inflation dynamics primarily to traditional supply and demand forces. In contrast, at the posterior mean, shortage shocks become meaningful contributors to this inflation episode.

6 Conclusions

This paper introduces a novel monthly newspaper-based shortage index that covers the period from 1900 through the present. The index captures the intensity of shortages across sectors—industry, labor, food and energy—and exhibits pronounced spikes during periods of economic turmoil, such as the world wars, the oil crises of the 1970s, and the COVID-19 pandemic. Validation exercises confirm that the index accurately measures shortages and correlates strongly with other indicators of supply constraints. The extensive historical coverage of the index makes it a valuable tool for understanding long-term economic dynamics.

Predictive regressions demonstrate that, throughout history, increases in the shortage index predict persistently higher inflation and lower economic activity. Furthermore, the relationship between inflation and shortages appears to be stronger during the COVID-19 pandemic. Structural VAR analysis further decomposes shortage movements into components arising from traditional business cycle shocks and exogenous innovations. Our results indicate that shortages cannot be explained solely by conventional demand, supply, commodity, and monetary shocks; surprise shortage innovations contribute importantly to inflation. Historically, the 1950s saw shortages driven mainly by demand factors, while the 1970s witnessed unusual shortage shocks from the OPEC oil embargo. The COVID-19 period reflects a mix of supply and demand factors that led to a sustained inflationary impact, with supply shocks—including shocks to shortages—playing a predominant role.

Our findings challenge a fundamental assumption that pervades much of modern macroeconomic theory. Traditional models either ignore shortages entirely or dismiss them as transitory symptoms of sluggish price adjustment—a temporary inconvenience on the path to market clearing. This treatment is inadequate. Shortages are neither fleeting nor trivial. They are persistent, measurable phenomena that recur throughout economic history and exert substantial, independent effects on inflation and real activity. Recognizing shortages as an empirically relevant state variable—distinct from conventional supply and demand disturbances—is essential for understanding the full dynamics of aggregate fluctuations. Our index provides researchers and policymakers with a systematic tool to measure, track, and analyze shortage episodes across more than a century of U.S. economic history. By establishing shortages as a first-order feature of the macroeconomic landscape, we open the door to testing the predictions of richer theoretical models that explicitly incorporate capacity constraints, rationing, and

non-price allocation mechanisms. We hope that future research will use this index to deepen our understanding of when shortages emerge, how they propagate through the economy, and what policy interventions can most effectively mitigate their costs.

Although our analysis relies on linear models to estimate the effects of shocks to the shortage index, future research could extend this framework to account for nonlinear dynamics using both data and theoretical models. Recent work has highlighted how supply chain disruptions and input bottlenecks can amplify shocks through production complementarities and rigidities (Bonadio et al., 2021; Baqaee and Farhi, 2022). Similarly, shortages in the form of binding capacity constraints and pricing frictions can lead to non-linear inflation responses (Guerrieri et al., 2022; Di Giovanni et al., 2023).

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Table 1: Search Query for the Construction of the Shortage Index

Search Name	**Search Query**	**Peak Month**
Energy Shortages	(*shortages* N/5 *energy*) AND *economics*	Dec-1973
Food Shortages	(*shortages* N/5 *food*) AND *economics*	Mar-1943
Industry Shortages	(*shortages* N/5 *industry*) AND *economics*	Aug-1942
Labor Shortages	(*shortages* N/5 *labor*) AND *economics*	Jan-2022
Articles	*articles*	–

Topic Sets	**Components**
shortages	`shortage* OR bottleneck* OR scarcit* OR rationing*`
energy	`oil OR gas OR coal OR fuel OR fuels OR gasoline OR energy OR heating OR petroleum OR electricity OR refinery OR pipeline OR petrol`
food	`food OR wheat OR meat OR milk OR crop OR crops OR grain OR farm OR agriculture OR famine OR feed OR farmer OR farmers OR water OR fertilizer OR drought`
industry	`steel OR copper OR iron OR metal* OR automotive OR textile OR machinery OR equipment OR transportation OR railway OR airline OR freight OR shipping OR transit OR deliveries OR shipment* OR ships OR chip* OR semiconductor* OR infrastructure OR materials OR distribution OR car OR cars OR parts OR goods OR material OR auto OR computer OR 'supply chain' OR components`
labor	`labor OR workers OR job* OR work OR employment OR manpower OR worker OR staff OR professional* OR technician* OR staffing OR skills OR workforce OR personnel OR strike* OR union*`
economics	`economic OR industr* OR production OR manufactur* OR economy OR trade OR commerce OR business OR budget OR tax OR fiscal OR corporation OR market OR price OR capacity OR company OR demand OR sales OR factory OR wages OR suppl*`
articles	`the AND be AND to AND of AND and AND at AND in`

Note: The truncation character (*) denotes a search encompassing all possible endings of a word, e.g. "scarcit*" includes "scarcity" and "scarcities." "AND" and "OR" are logical operators, N/5 denotes a proximity operator requiring words appearing within five words from each other.

Table 2: Largest Shortage Spikes, 1900-2024

Month	Index	Surprise (st.dev.)	Event
Jan-1903	174	2.84	Nationwide coal shortages
Dec-1906	191	2.58	Shortage of coal and freight cars in Midwest
Nov-1916	235	2.54	Nationwide coal shortages
Jan-1918	578	7.63	Fuel and coal shortages
Dec-1919	300	2.75	Fuel and coal shortages due to war and strikes
Jul-1920	317	2.32	Freight car shortage affects coal and steel transportation
Aug-1922	281	3.24	Coal shortage due to strikes
Aug-1930	93	1.28	Drought leads to food and water shortages
Jul-1934	134	1.70	Strike by Teamsters unions in the West Coast
Sep-1939	152	1.81	Steel shortage due to the beginning of World War II
Aug-1941	516	3.30	War-related energy, materials and labor shortages
Jan-1943	1085	4.87	War-related oil, labor and food shortages
Jan-1945	563	4.17	War-related widespread shortages
Aug-1945	556	3.46	Labor shortages at the end of war
May-1946	589	4.31	Strikes by coal workers and fuel shortages
Jan-1948	460	4.01	Metal, fuel and food shortages
Feb-1950	226	2.30	Coal shortages amid strikes
Jan-1951	286	2.44	Labor shortages due to demand from defense industries
Jan-1952	271	2.87	Nationwide and worldwide shortages
Dec-1956	172	1.80	Oil shortages due to Suez Crisis
Dec-1973	1085	11.28	Gasoline shortages due to 1973 oil crisis
Jan-1975	267	2.02	Concerns about gasoline rationing
Feb-1977	422	6.81	Carter's appeal on energy conservation
Feb-1978	158	1.59	Concerns about energy shortages
May-1979	579	4.81	Concerns about energy shortages
Aug-1981	153	1.18	Gasoline shortages due to 1979 oil crisis
Aug-1990	160	1.47	Concerns about energy shortages
Apr-2020	297	4.58	Medical shortages due to COVID-19 pandemic
Jan-2022	554	2.69	Labor shortages
Oct-2023	213	1.20	United Auto Workers strike

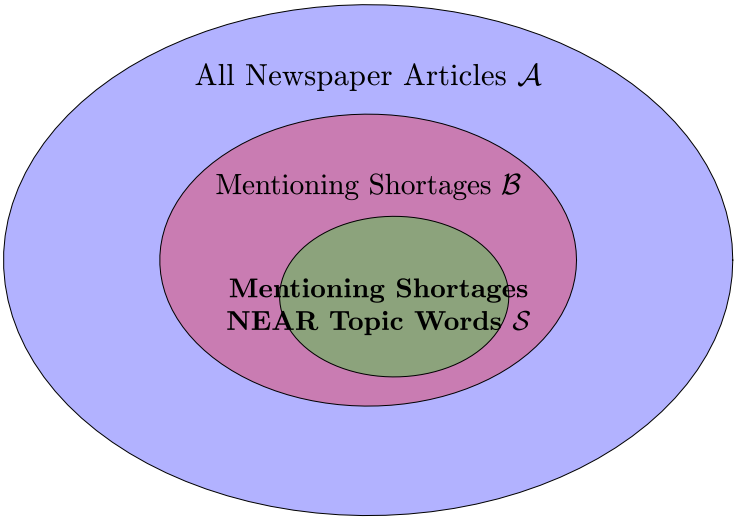
Note: The table lists the 30 largest spikes in the shortage index. For this table, the spikes are identified using the residuals of an autoregression and a condition on local maxima described in footnote 4.

Table 3: Validation of the Shortage Index

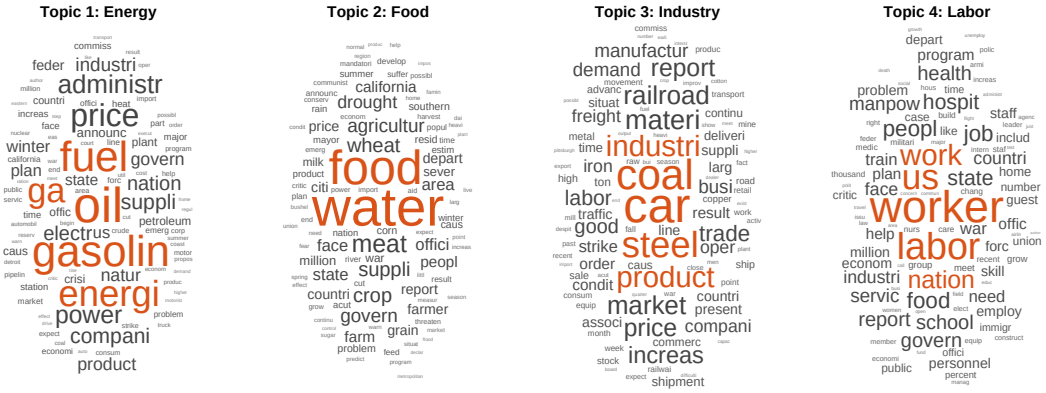
Set	**Share of Total Newspaper Articles in Set**	**Validation Sample**	**Articles Mentioning Actual Shortages**	**Type I Error**	**Type II Error**
Shortages NEAR Topic Words ($\mathcal{S}$)	1.58%	872	817	6.30%	–
Not Shortages ($\mathcal{A} \setminus \mathcal{S}$)	98.42%	298	1	–	0.33%
All Shortages ($\mathcal{B}$)	2.93%	334	284	14.97%	–

Note: Validation of the shortage index using a sample of newspaper articles used to construct the index, denoted by the set Shortages NEAR Topic Words ($\mathcal{S}$), a sample of articles excluded from the construction of the index (denoted by the set $\mathcal{A} \setminus \mathcal{S}$), and a sample of articles mentioning All Shortages ($\mathcal{B}$).

Figure 1: Grouping of Newspaper Articles for the Construction of the Shortages Index



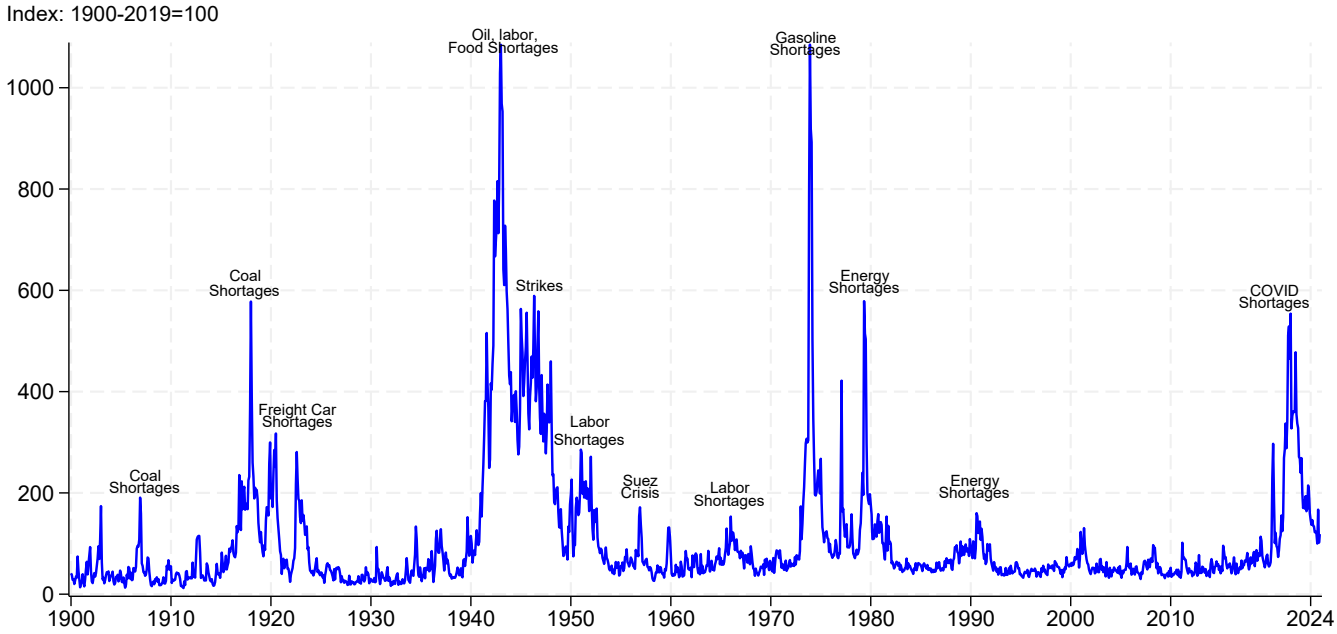
(a) Grouping of Newspaper Articles



(b) Word Clouds and Topic Classification of Articles

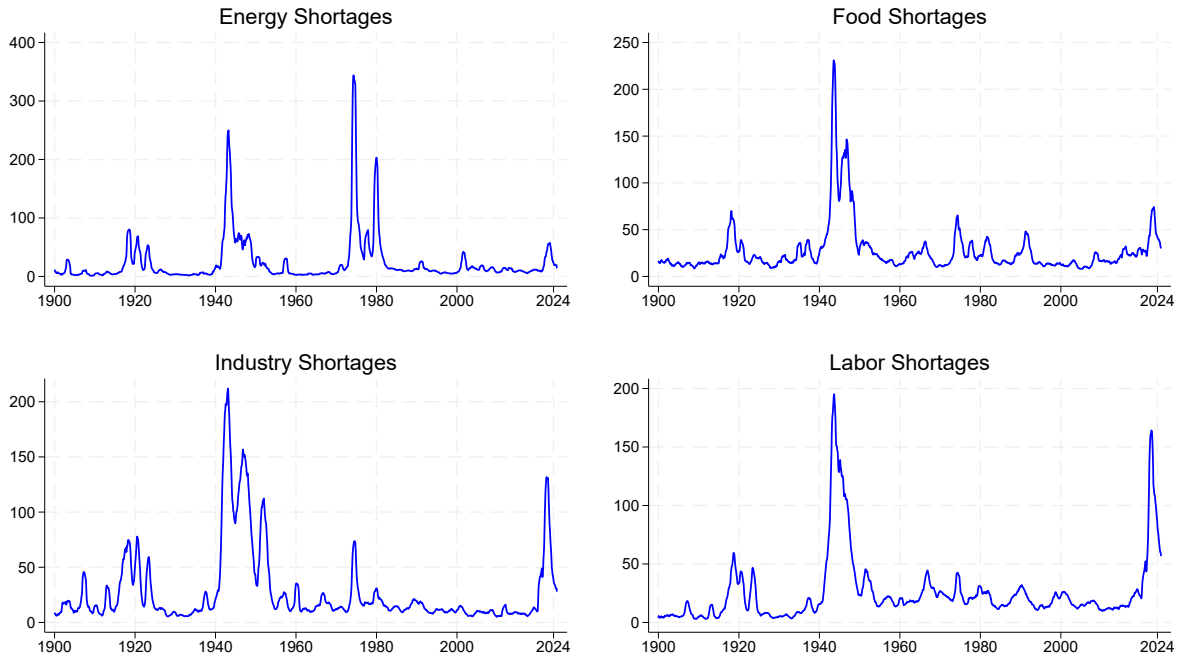
Note: Panel (a) shows a diagram representing the set of articles used in the construction of the index. Panel (b) shows word clouds associated with the 4 most influential topics chosen by Latent Dirichlet Allocation (LDA) analysis on a large set of articles.

Figure 2: The Shortage Index

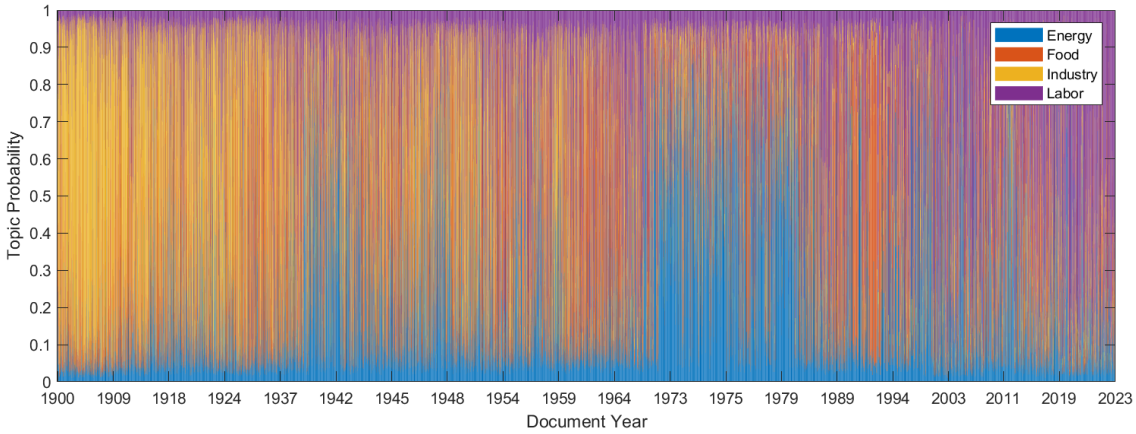


Note: The figure plots the shortage index from January 1900 through December 2024. Updated charts and data can be found at https://www.matteoiacoviello.com/shortages.html.

Figure 3: Decomposition of the Shortage Index by Category



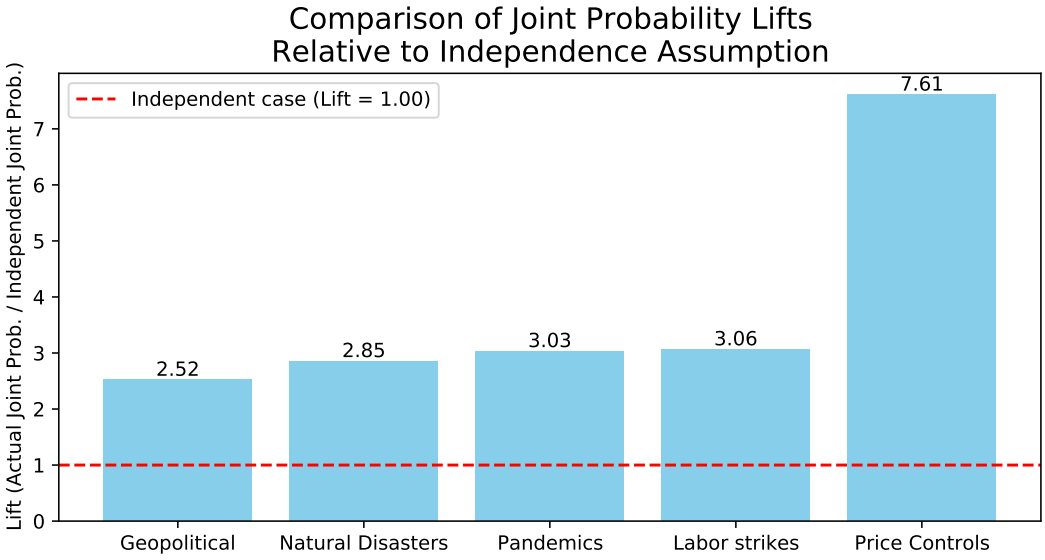
(a) Decomposition of the Shortage Index by Category



(b) Topic Classification for the Index

Note: Panel (a) shows shortage indexes by category, scaled so that the sum of the four indexes adds up to the total index shown in Figure 2. Panel (b) shows the evolution over time of the topic classification by a Latent Dirichlet Allocation algorithm for a large sample of articles used in the construction of the index.

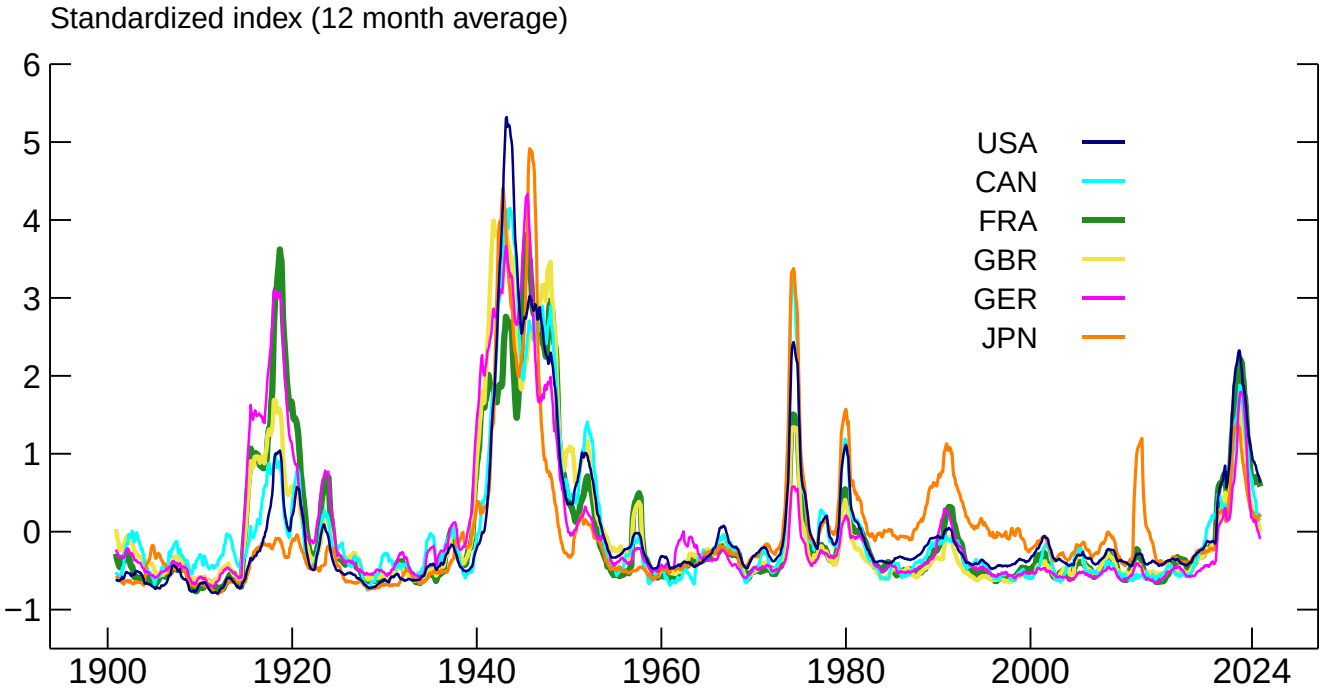
Figure 4: Phenomena Associated with Shortages



Category	***P(Shortage)***	***P(Category)***	***P(Joint)***	***P(Indep.)***
Geopolitical	1.27%	2.63%	0.084%	0.033%
Natural Disasters	1.27%	2.85%	0.103%	0.036%
Pandemics	1.27%	1.23%	0.047%	0.016%
Labor strikes	1.27%	2.69%	0.104%	0.034%
Price Controls	1.27%	0.80%	0.077%	0.010%

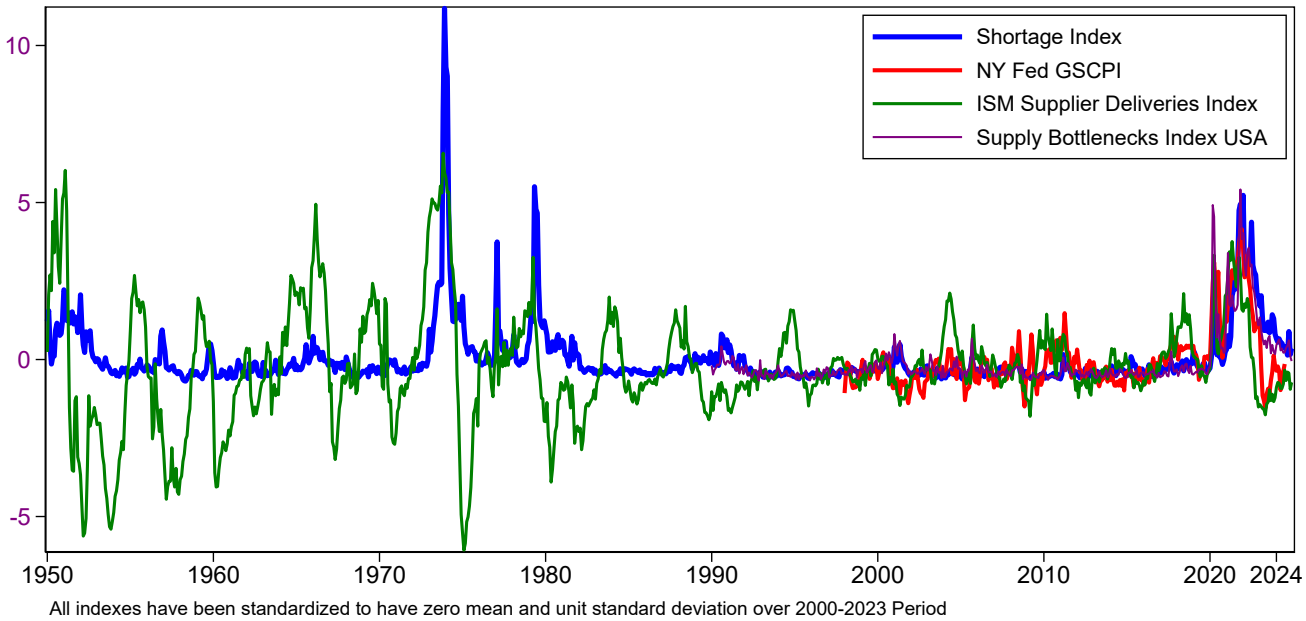
Note: The figure plots lift ratios, calculated as the ratio between the realized joint probability of events in 5 categories being mentioned in newspaper articles alongside shortages, and the probability that would be expected if these events and shortages were independent. The higher the lift ratio, the stronger the association between events and shortages. In the table, *P(Shortage)* is the frequency of news articles mentioning shortages; *P(Category)* is the frequency of news articles mentioning the specific category in each row; *P(Joint)* is the frequency of news articles mentioning both shortages and the specific category in each row; and *P(Indep.)* is the hypothetical joint probability under the independence assumption.

Figure 5: Shortage Index across Countries



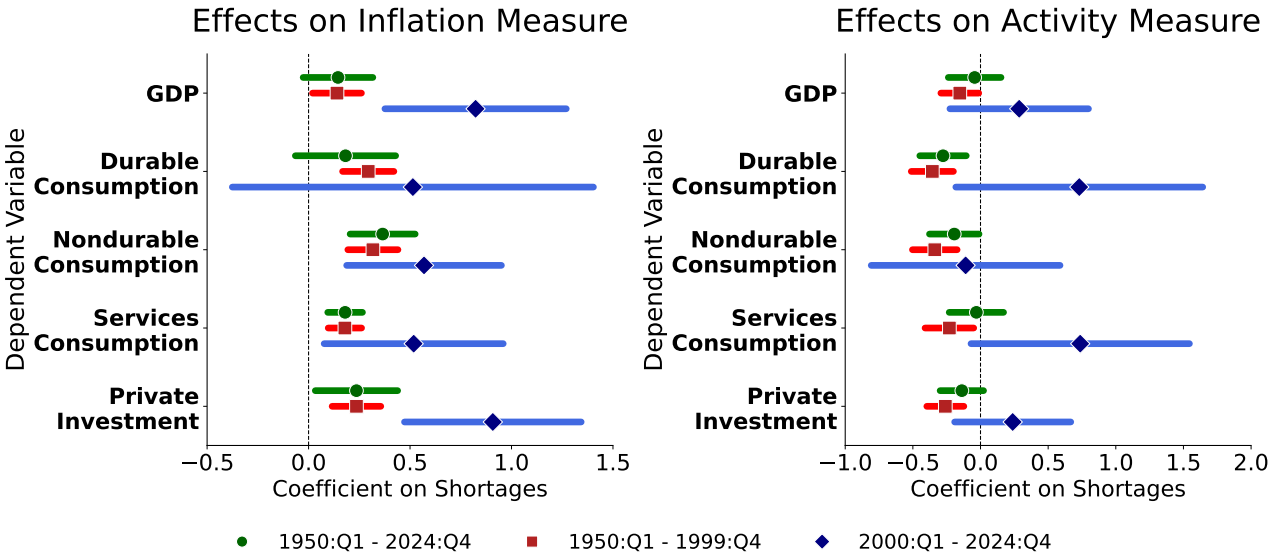
Note: The figure shows the shortage index for the United States, Canada, France, Germany, the United Kingdom, and Japan. The country indexes are proportional to the share of newspaper articles mentioning shortages and the name of a foreign country and/or one of its major cities. Indexes are standardized to have mean zero and unit standard deviation over the period 1900 through 2024.

Figure 6: Comparison to Other Supply Constraints Measures



Note: The figure compares the shortage index to alternative measures of supply constraints. The GSCPI is the New York Fed Global Supply Chain Pressure index (Benigno et al., 2022), measuring global supply chain conditions using data on both manufacturing and transportation costs. The ISM Supplier Deliveries Index is computed as the share of respondents reporting longer delivery times plus half the share of respondents reporting no change in delivery times. U.S. The Supply Bottlenecks Index is from Burriel et al. (2023).

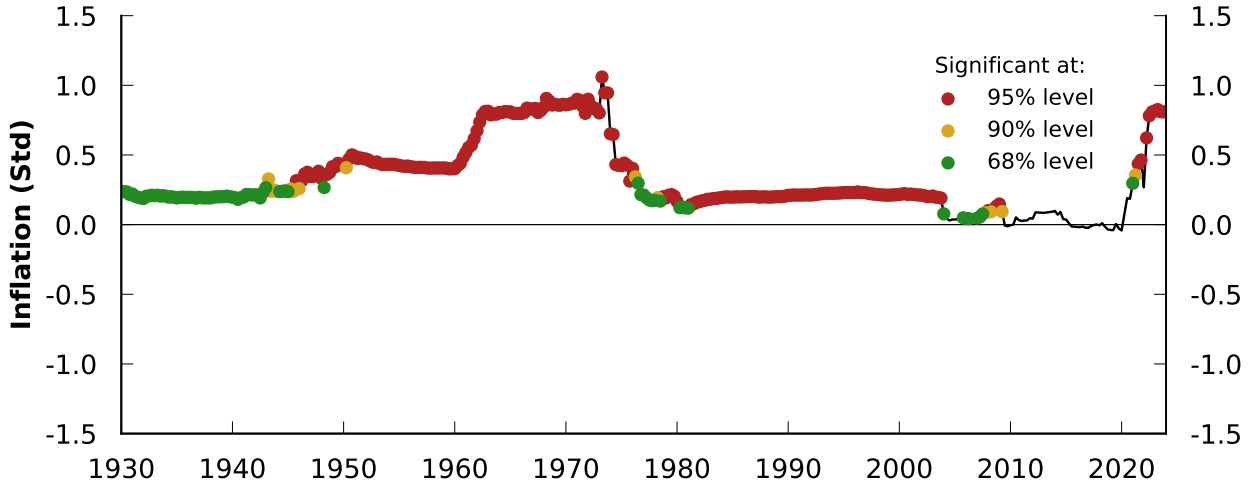
Figure 7: Relationship between Shortages and Inflation and Activity 4-Quarters Ahead



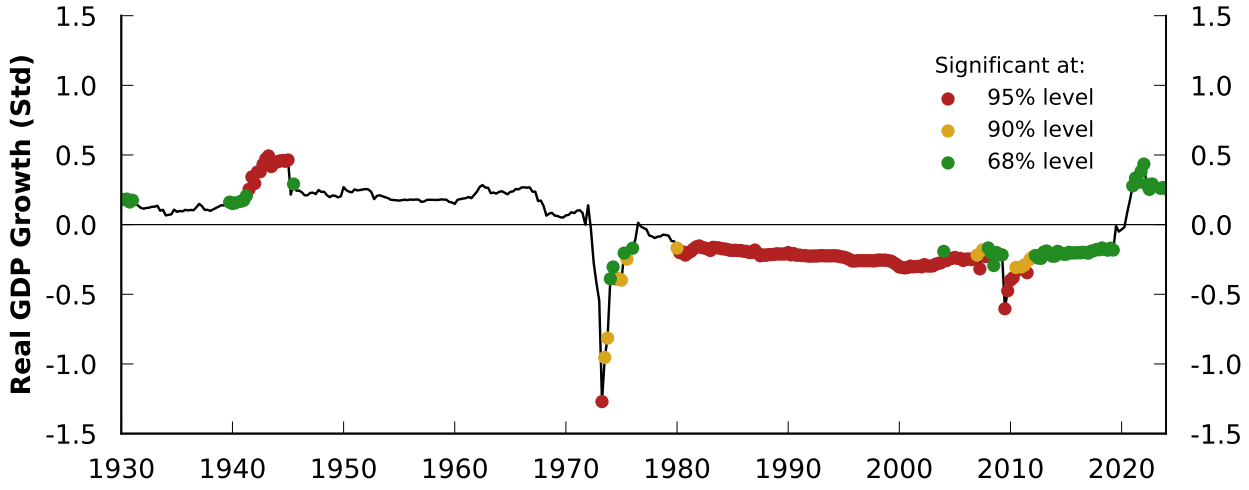
Note: The figure reports standardized coefficients of predictive regressions of inflation and economic activity on the shortage index estimated on quarterly data. The dependent variable for each regression is the log difference between $t+4$ and t of the variable listed in each row (the price deflator for inflation and real quantities for growth). Each regression includes as controls the quarterly changes of the dependent variable and the associated economic indicator or price deflator, contemporaneously and with 3 lags. Data are quarterly. The full sample runs from 1950:Q1 to 2024:Q4. We also partition the sample into 2 periods: 1950:Q1 to 1999:Q4, and 2000:Q1 to 2024:Q4. Heteroskedasticity consistent 90 percent confidence intervals are reported using shaded horizontal bars.

Figure 8: Effect of Shortages on Inflation and Real GDP Growth (30-Year Window)

(a) One-Year Ahead Inflation

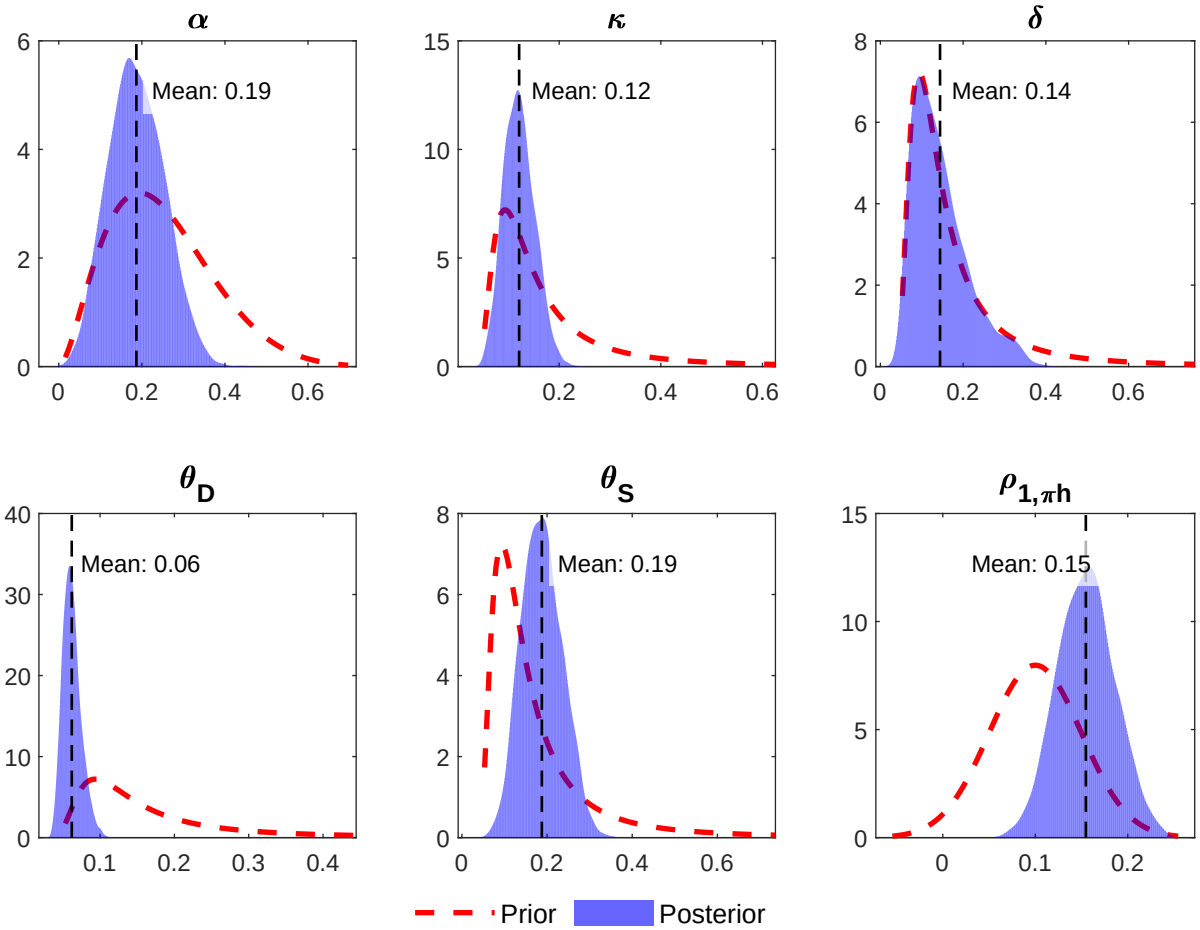


(b) One-Year Ahead Real GDP Growth



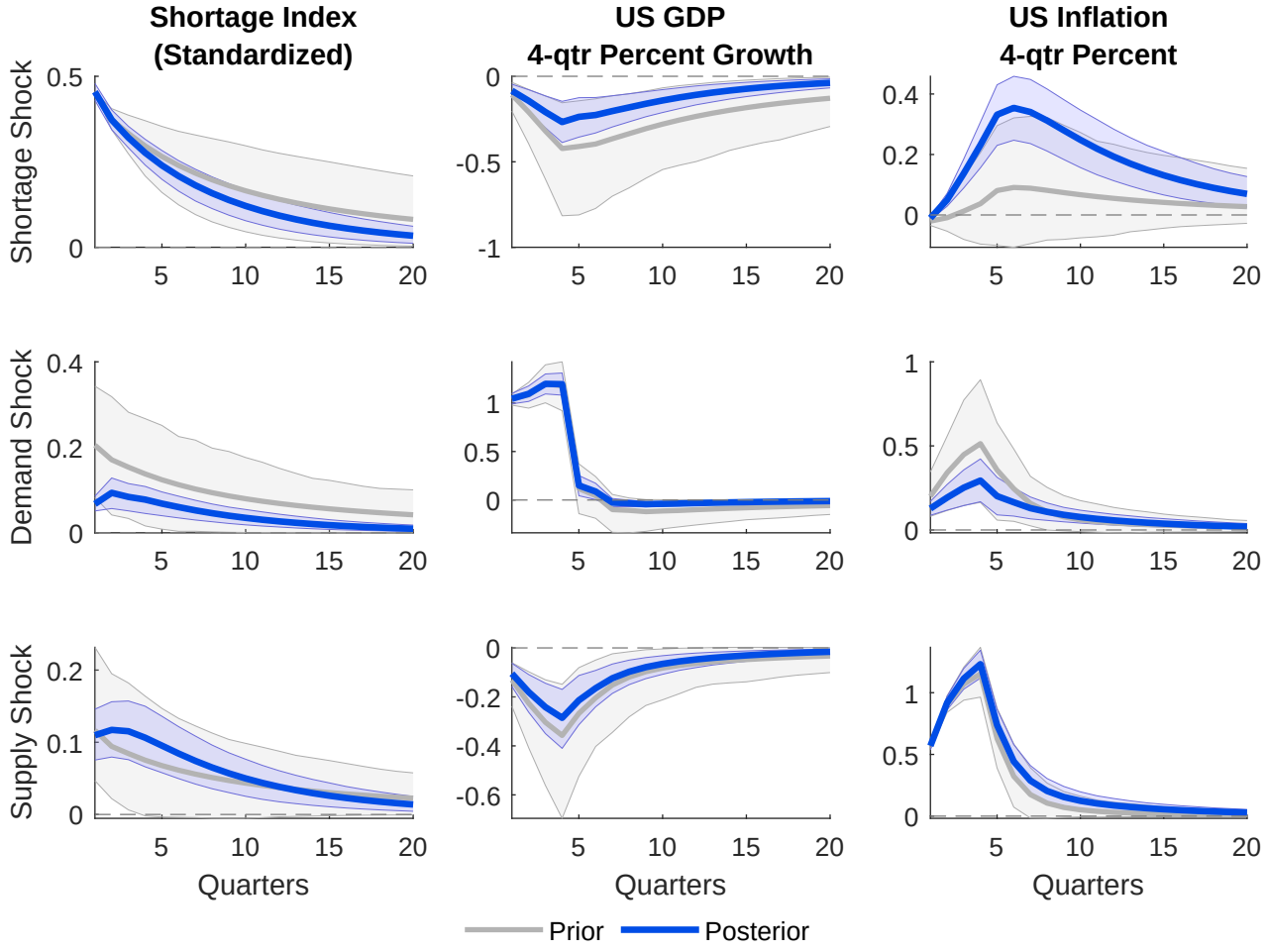
Note: The black line shows the time-varying estimated effect of shortages on inflation (panel (a)) and real GDP growth (panel (b)). The estimates are based on regressions using rolling 30-year windows. In each regression, the dependent variable is the 4-quarter ahead difference in log real GDP per capita or the 4-quarter ahead difference in log GDP deflator. On the right-hand side, the main explanatory variable is our shortage index. As controls, we include the 1-quarter change in both log real GDP per capita and log GDP deflator, in quarter t plus 3 lags. Significance is computed based on heteroskedasticity consistent standard errors.

Figure 9: Prior and Posterior Densities of Structural VAR Parameters



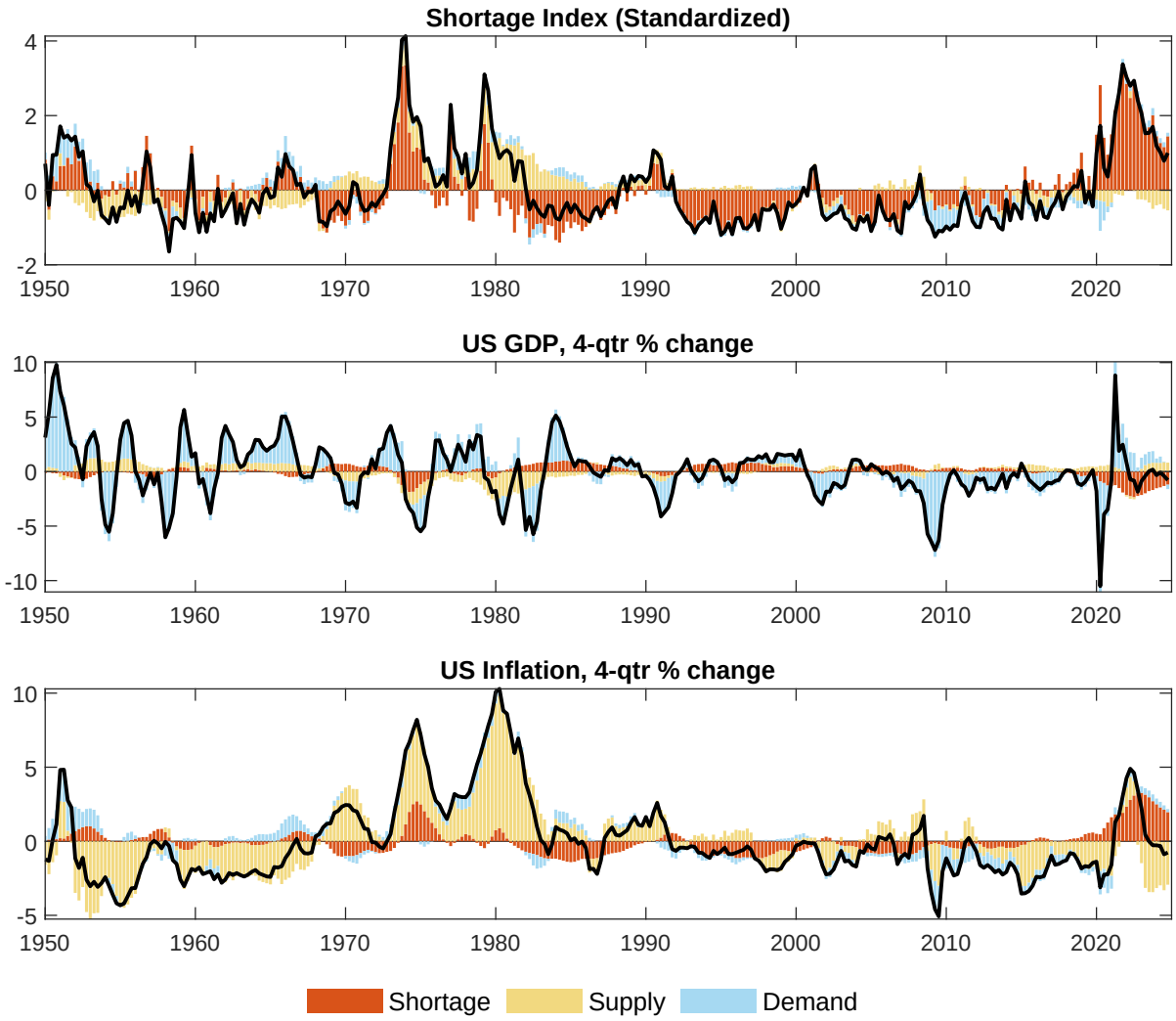
Note: Prior and posterior densities of selected parameters of the structural VAR model. See Section 5 for additional details and the Appendix for the full list of prior and posterior moments of the model parameters.

Figure 10: Effects of Shortages on US Activity and Inflation in the Structural VAR Model



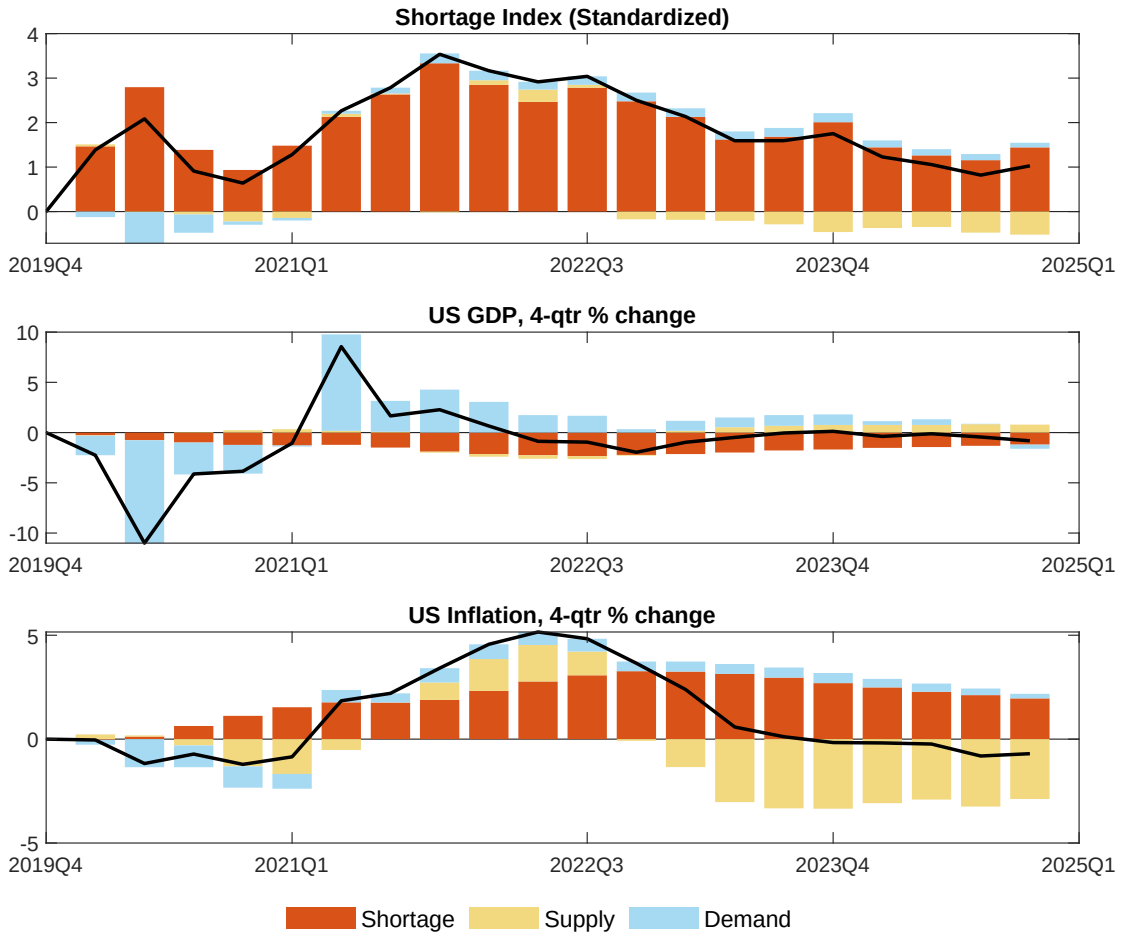
Note: Impulse responses to structural shocks of size one standard deviation estimated using the structural VAR model described in Section 5. Gray and blue lines denote the responses at the prior and posterior mean of the parameters, respectively. Gray and blue shaded areas denote prior and posterior 80 percent confidence intervals, respectively.

Figure 11: Historical Decomposition from the Structural VAR Model: Full Sample



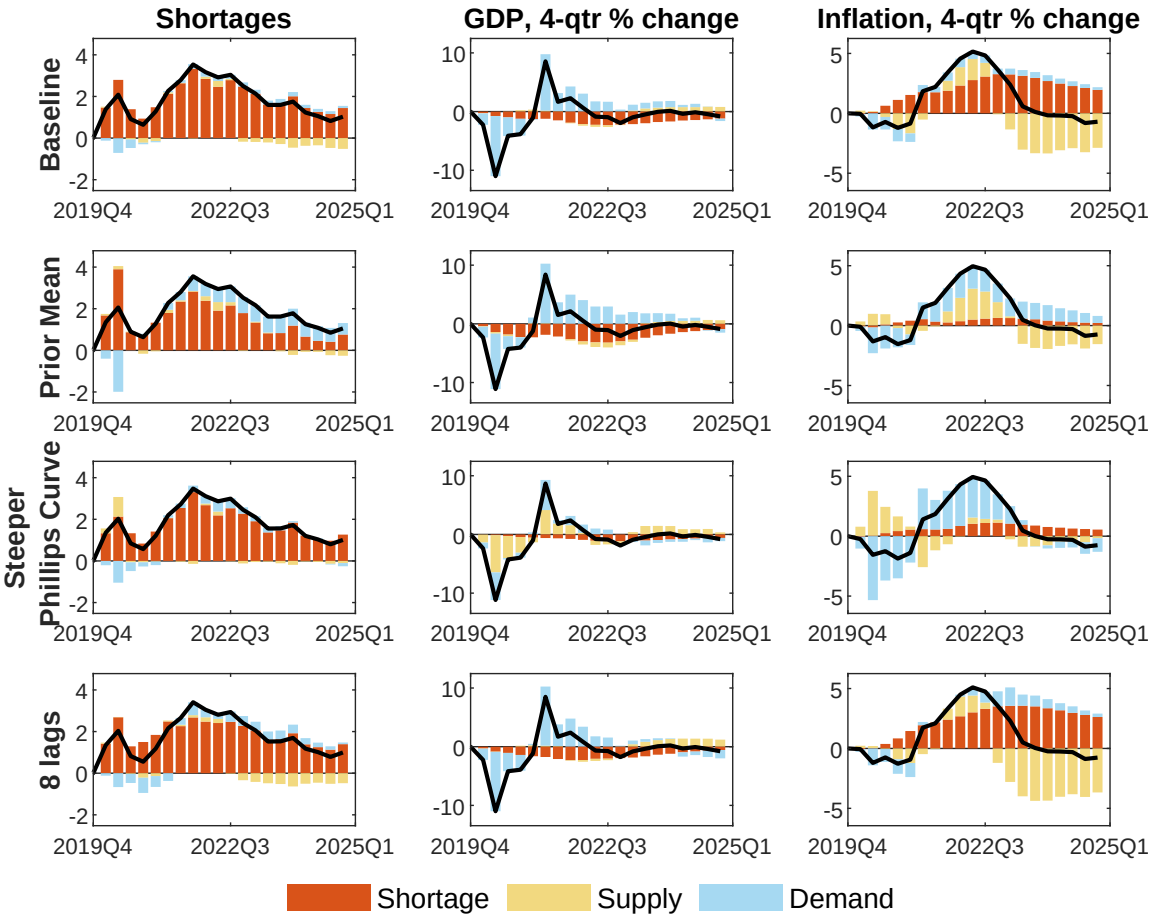
Note: This figure displays the historical contributions of structural shocks to the evolution of the model's endogenous variables. All variables are expressed as deviations from their sample means.

Figure 12: Historical Decomposition from the Structural VAR Model: 2020-2024



Note: This figure displays the contributions of structural shocks to the evolution of the model's endogenous variables since 2020. All variables are expressed in deviation from their 2019:Q4 value, and any shocks that occurred before 2019:Q4 are set equal to zero.

Figure 13: Historical Decomposition from the Structural VAR Model: 2020-2024
Alternative Models



Note: This figure displays the contributions of structural shocks to the evolution of the model's endogenous variables since 2020. All variables are expressed in deviation from their 2019:Q4 value, and any shocks that occurred before 2019:Q4 are set equal to zero. Starting from the top, the first panel reports results from the baseline model at the posterior mean; the second panel shows results from the a model estimated with all parameters except the shocks standard deviations held at the prior mean; the third panel shows results for an alternative model featuring a steeper Phillips curve; the bottom panel shows results for a version of the model augmented to include eight lags.

Appendix for: Measuring Shortages since 1900

A Data Sources

This Section describes in detail the data used in the paper in addition to the shortage indexes presented throughout the paper and available at https://www.matteoiacoviello.com/shortages.html.

The shortage indexes presented in the paper are constructed and analyzed running search queries using ProQuest Global Newsstream Collection, ProQuest Historical Newspapers, and the textual analytics tools of ProQuest TDM Studio (see https://tdmstudio.proquest.com/home).

Section 3 and the Appendix describe the results of regressions of selected price and wage indicators on the categorical shortage indexes. In addition to the shortages indexes, the series we use in these regressions and the corresponding data sources are:

- Wages, measured by BLS Average Hourly Earnings in Total Private Industries (Haver mnemonics: LETPRIVA@USECON);
- Producer Prices, measured by PPI: Intermediate Demand Processed Goods, seasonally adjusted (Haver mnemonics: PC1@USECON);
- CPI: Energy (Haver mnemonics: PCUSEN@USECON);
- CPI: Food (Haver mnemonics: PCUFON@USECON).

Section 3 also compares our shortage index with alternative measures of supply constraints. Data sources are as follows:

- The Global Supply Chain Pressure Index (GSCPI) published by the Federal Reserve Bank of New York and described in Benigno et al. (2022) (Haver mnemonics: GSCPI@SURVEYS);
- The ISM Supplier Deliveries Index (SDI) produced by the Institute for Supply Management (Haver mnemonics: NAPMVDIN@SURVEYS);
- U.S. Supply Bottlenecks Index (SBI) from Burriel et al. (2023).

The predictive regressions in Section 4 use data on real GDP, personal consumption expenditures (durables, non-durables, and services), and private fixed investment and their associated deflators. Data for all variables cover the period 1950:Q1-2024:Q4 and are taken from the Bureau of Economic Analysis National Income and Product Account Historical Tables. Variables are expressed in per capita terms dividing by total population from the U.S. Census Bureau, taken from Haver (Total Population: All Ages including Armed Forces Overseas;

Haver mnemonics: POPT@USECON). The population series starts in 1952 and is backcast to 1950 using population data from Ramey and Zubairy (2018).

In Section 4, we run the predictive regressions for GDP and its deflator on a rolling sample for the period 1900:Q1-2024:Q4. To this end, we backcast the GDP, GDP deflator, and population data described in the previous paragraph using the growth rates of the corresponding series in Ramey and Zubairy (2018), which are available for the earlier period.

The regressions in Section 4 also use as controls the following series:

- Oil Prices, measured by Spot Oil Price: West Texas Intermediate. (Haver mnemonics: PZTEXP@USECON);
- Commodity Prices, measured by the S&P Goldman Sachs Commodity Nearby Index (Haver mnemonics: GSCI@USECON);
- Oil Prices and Commodity Prices are expressed in real terms by dividing for the Seasonally Adjusted PCE Chain Price Index (Haver mnemonics: JCBM@USECON);
- Wages, measured by BLS Average Hourly Earnings in Total Private Industries from the BLS (Haver mnemonics: LETPRIVA@USECON);
- 10-year Inflation Expectations from the Federal Reserve Bank of Cleveland, retrieved from FRED (Fred mnemonics: EXPINF10YR).

The VAR analysis of Section 5 uses the following variables:

- U.S. real GDP, from the Bureau of Economic Analysis;
- CPI for All Items less Food and Energy (Haver mnemonics: PCUSLFEN@USECON);

B Additional Details of Audit Conducted with Claude AI

We used the Claude AI Assistant to help with auditing the index. Claude was instructed to return a table of results, coding articles as 1 (shortage mentioned), 0 (shortage not mentioned), or 99 (unsure whether the existence of shortages was mentioned). In addition to the classification, Claude was asked to provide a brief explanation for each snippet's coding.

Before initiating the classification, we provided Claude with examples of how we would code the snippets and ensured that the training sample included false positives, such as mentions of the lack or end of shortages. The specific prompt was as follows:

"I give you 872 snippets of text, each about 110 characters long. For each of them, can you tell me whether they mention current or prospective shortages, rationing, scarcity, or bottlenecks related to goods, labor, materials, food, or water? Just return a table with yes=1, no=0, unsure=99, and a brief explanation. For instance:

- Article 1 mentions prospective shortages since it states that steel shortages will prevail in the near future, so it is coded as 1.
- Article 2 says steel shortages caused a plant closure, so it is coded as 1.
- Article 3 says the shortage of cars is crimping coal production, so it is coded as 1.
- Article 4 mentions the shortage of cars, so it is coded as 1.
- Article 10 mentions a shortage of workspace, so it is not related to a work shortage and is coded as 0.
- Article 329 says no shortage of cars has been experienced, so it is coded as 0."

Although using AI for validation is not foolproof, we found that Claude performed as well as, if not better than, a human classifier. For example, Claude demonstrated an ability to extrapolate the context of a particular sentence to a country or person. In one instance, the snippet "Economy may be slowing, but Lowe is banking on labor shortages gradually leading to an increase..." was classified as 1 by Claude, with the explanation: "Reserve Bank [of Australia] expecting labor shortages to lead to wage growth."

The results of the audit are in Table 3. Out of 872 articles belonging to the set $\mathcal{S}$, only 6.3 percent were deemed by Claude as false positives. Claude classified the snippet "...says that while there is not necessarily a shortage of people wanting to work in management" as 0 with the explanation "No shortage of people wanting to work in management." Similarly, the snippet "a motive for mr. newt gingrich's knife job, had no shortage of conspiracy theories, most leading to the..." was classified as 0 with the explanation "Speaker's ouster sparked many conspiracy theories but not actual shortages." In some cases, Claude classified as 0 articles that we would have probably classified as 1. For instance, the snippet "canada's action today in temporarily suspending meat rationing" was classified as 0 since Claude gave more weight to the temporary suspension of the rationing rather than its existence. Of note, Claude classified only one snippet as "unsure," and we found it had good reason to do so: the snippet was sampled from a short article in the Chicago Tribune in 1929 that reported short snippets of information on miscellaneous news items, a common practice in early 20th-century journalism.

The snippet mentioned “Food shortage in Batanes islands. (Neither do we.)...” and featured incomplete comparison, lack of context, grammatical mismatch, and ambiguous meaning.

C Additional Empirical Results

Validation of the Categorical Shortage Indexes

Table A.1 reports the coefficients from regressions where selected price and wage measures are regressed on the four categorical shortage indexes for industrial products, labor, energy and food. The dependent variables in these regressions are prices indexes that are more likely to be impact by shortages in these four categories: the 3-month ahead log difference of the Producer Price Index (PPI) for processed goods for intermediate demand, the Consumer Price Index (CPI) for energy, CPI for food, and the 12-month ahead log change in average hourly earnings (total private). Each regression includes three lags of the monthly changes in the dependent variable and one lag of the log change in industrial production.[1]

When the four categorical indexes are included one at a time, we find that the industry shortage sub-index predicts PPI materials inflation, the labor shortage index predicts earnings growth, and the energy and food shortage indexes predict energy CPI inflation and food CPI inflation, respectively. In other words, each index forecasts the relevant prices in the sectors most likely to be affected by that specific shortage. When all four indexes are included simultaneously, these results largely hold, with one exception: the food shortage index no longer predicts CPI food inflation. In addition, energy shortages predict PPI materials inflation and (weakly) wage growth.

Additional Results from Predictive Regressions

We explore the relationship between shortages, inflation, and economic activity using the following predictive regression:

$$\Delta Y_{t+h} = \alpha + \beta\, SHORTAGE_t + \sum_{i=0}^{p} \boldsymbol{\gamma}_i' \mathbf{X}_{t-i} + \varepsilon_{t+h} \tag{7}$$

where

$$\Delta Y_{t+h} = \frac{400}{h} \ln\left(\frac{Y_{t+h}}{Y_t}\right)$$

represents the annualized log change of a variable of interest Y_t between period t and forecast horizon h, and $SHORTAGE_t$ denotes the level of the shortage index. The vector $\mathbf{X}$ contains control variables. We use quarterly data from 1950 through 2024 from the National Income and Product Accounts (NIPA) for real per-capita GDP, personal consumption expenditures, and private fixed investment. Inflation for each category is measured using the associated price deflator. For GDP and its price deflator, we extend the sample back to 1900 using data from Ramey and Zubairy (2018). Data on total population also come from Ramey and Zubairy (2018), extended through 2024 using the POP series from FRED.

For each price and economic activity indicator, we estimate equation (7) by ordinary least squares, with standard errors calculated following Newey and West (1987). As control variables, we include quarterly changes of the dependent variable and the corresponding economic indicator or price deflator, both contemporaneously and with three lags. For example,

[1] We use the 12-month change for earnings because earnings adjust more slowly than commodity prices or materials. Our results are similar when using the 6-month change in earnings.

the regression for real GDP growth includes contemporaneous and lagged values of both real GDP growth and inflation (measured by the log change in the GDP deflator).

Figures A.1 and A.2 report results for the one- and eight-quarter horizons, respectively. For ease of comparison, we report standardized estimates of the coefficient β. A standardized coefficient represents the change in standard deviation units in the dependent variable following a one-standard-deviation change in the explanatory variable.

Does the Shortage Index Contain Information Beyond Traditional Indicators?

One concern discussed in the paper is whether the shortage index contains information beyond that captured by other macroeconomic variables. We address this issue by augmenting our baseline regression with additional controls, including oil prices, commodity prices, wage growth, and inflation expectations. The inclusion of these controls only slightly attenuates the effect of shortages on inflation and activity, but the coefficients remain in large part statistically significant. See Figure A.3 for details. These findings suggest that our shortage index contains additional information beyond that captured by traditional macroeconomic indicators.

VAR Results: Additional Detail

Table A.2 provides details of the prior distributions for all model parameters, along with their posterior mean, standard deviation, and 80 percent credible intervals.

Table A.1: Predictive Regressions with Categorical Shortage Indexes

	Wages		PPI Materials		CPI Energy		CPI Food	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Labor Shortages	0.06*	0.02		-0.14		0.04		-0.09
	(1.80)	(0.24)		(-1.20)		(0.40)		(-1.05)
Industry Shortages		0.03	0.22**	0.36***		0.14		0.25**
		(0.24)	(2.47)	(2.88)		(1.54)		(2.50)
Energy Shortages		0.17*		0.21***	0.26***	0.24***		0.11
		(1.80)		(3.98)	(5.42)	(3.94)		(1.59)
Food Shortages		-0.01		-0.18**		-0.11*	0.17***	0.06
		(-0.34)		(-2.57)		(-1.66)	(3.52)	(1.16)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	725	725	729	729	729	729	729	729

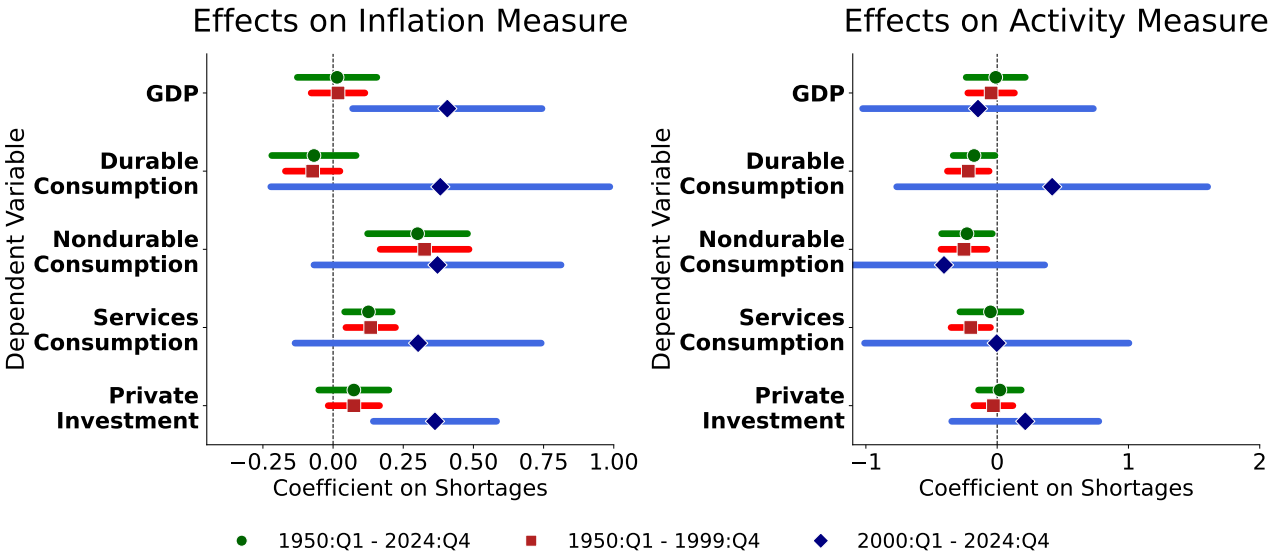
Note: The table reports standardized coefficients of predictive monthly regressions of selected prices and wages indicators on the categorical shortage indexes. The dependent variable for each regression is the 3-month log difference of the producer price index (PPI) for processed goods for intermediate demand, consumer price index (CPI) energy, CPI Food, and the 12-month log change in average hourly earnings (total private). Each regression includes as controls three lags of the monthly changes of the dependent variable and one lag of the log change in industrial production. The sample runs from 1964:M1 through 2024:M12. Heteroskedasticity and autocorrelation consistent t-statistics are reported in parentheses and computed according to Newey and West (1987). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.2: VAR Model: Estimated Parameters

		Prior		**Posterior**			
Parameter	Type	**Mean**	St. Dev	**Mean**	St. Dev	10%	90%
α	Beta	**0.25**	0.12	**0.19**	0.07	0.10	0.27
δ	Inv. Gamma	**0.20**	1.00	**0.14**	0.07	0.07	0.24
κ	Inv. Gamma	**0.20**	1.00	**0.12**	0.03	0.08	0.16
θ_D	Inv. Gamma	**0.20**	1.00	**0.06**	0.01	0.05	0.08
θ_S	Inv. Gamma	**0.20**	1.00	**0.19**	0.05	0.13	0.25
$\rho_{1,HH}$	Normal	**0.83**	0.06	**0.83**	0.04	0.78	0.88
$\rho_{1,H\pi}$	Normal	**0.00**	0.05	**0.05**	0.03	0.01	0.09
$\rho_{1,HY}$	Normal	**0.00**	0.05	**0.03**	0.02	0.00	0.06
$\rho_{1,\pi H}$	Normal	**0.10**	0.05	**0.15**	0.03	0.12	0.20
$\rho_{1,\pi\pi}$	Normal	**0.59**	0.06	**0.59**	0.04	0.54	0.65
$\rho_{1,\pi Y}$	Normal	**0.00**	0.05	**-0.03**	0.03	-0.06	0.01
$\rho_{1,YH}$	Normal	**0.00**	0.05	**0.06**	0.04	-0.00	0.11
$\rho_{1,Y\pi}$	Normal	**0.00**	0.05	**-0.00**	0.04	-0.06	0.06
$\rho_{1,YY}$	Normal	**0.10**	0.06	**0.07**	0.04	0.02	0.12
$\rho_{2,HH}$	Normal	**0.07**	0.06	**0.02**	0.04	-0.03	0.07
$\rho_{2,\pi\pi}$	Normal	**-0.05**	0.06	**-0.04**	0.04	-0.09	0.01
$\rho_{2,YY}$	Normal	**0.15**	0.06	**0.11**	0.04	0.06	0.16
σ_{ε_d}	Inv. Gamma	**1.11**	0.55	**1.08**	0.04	1.02	1.13
σ_{ε_h}	Inv. Gamma	**0.46**	0.23	**0.45**	0.02	0.43	0.48
σ_{ε_s}	Inv. Gamma	**0.63**	0.31	**0.58**	0.03	0.55	0.62

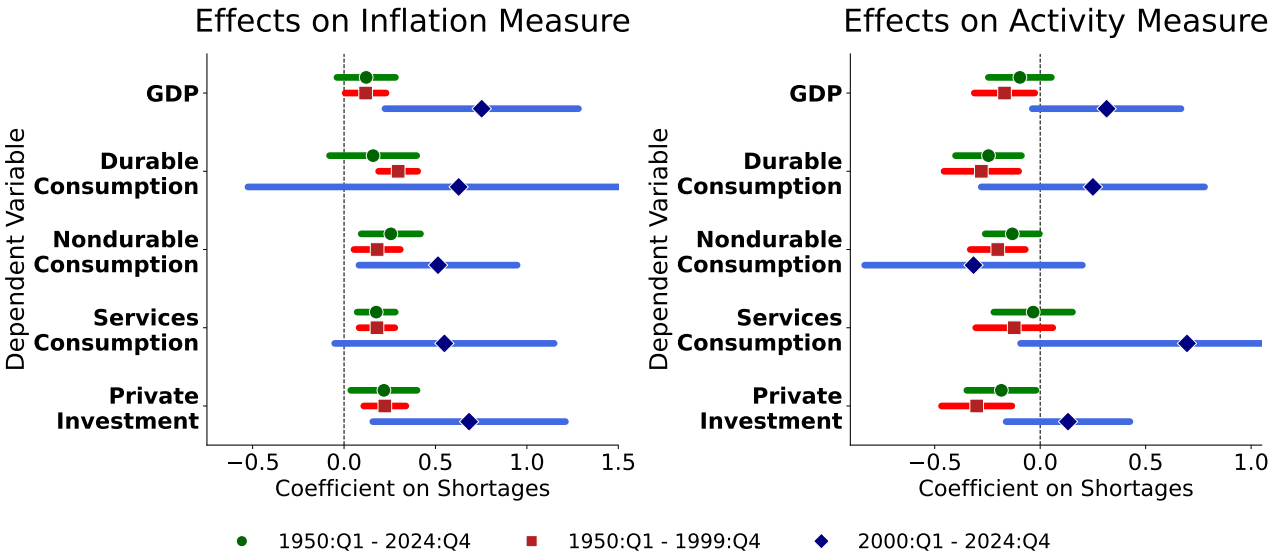
Note: The table shows estimated parameters of the model described in Section 5.

Figure A.1: Relationship between Shortages and Inflation and Activity One-Quarter Ahead



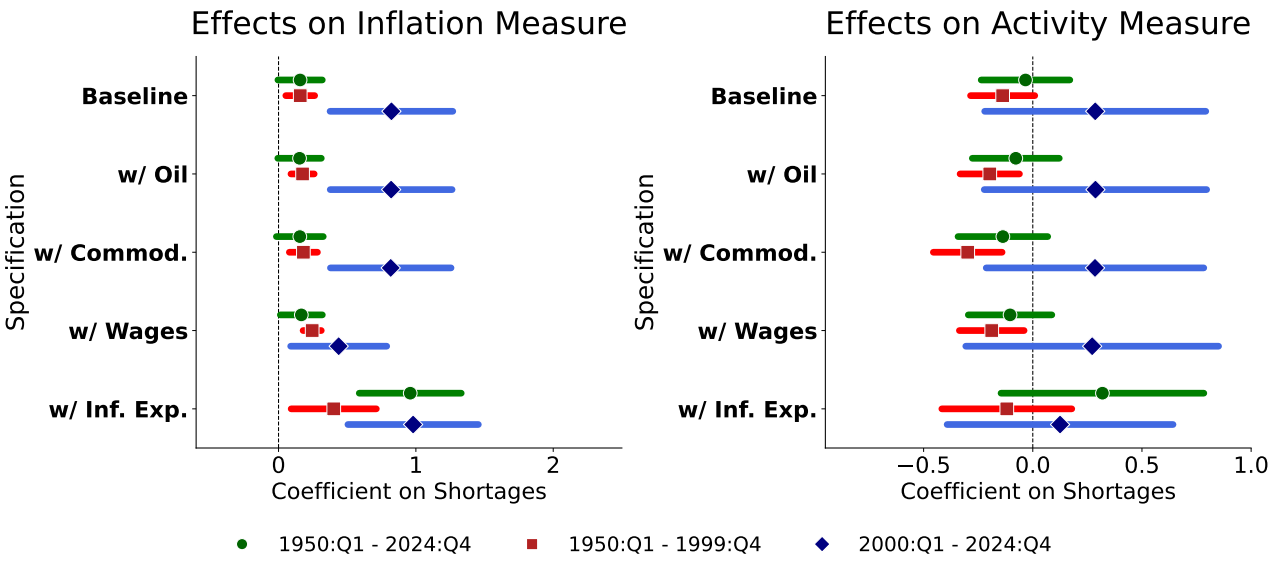
Note: The figure reports standardized coefficients of predictive regressions of inflation and economic activity on the shortage index estimated on quarterly data. The dependent variable for each regression is the log difference between $t+1$ and t of the variable listed in each row (the price deflator for inflation and real quantities for growth). Each regression includes as controls the quarterly changes of the dependent variable and the associated economic indicator or price deflator, contemporaneously and with three lags. Data are quarterly. The full sample runs from 1950:Q1 to 2024:Q4. We also partition the sample into two periods: 1950:Q1 to 1999:Q4, and 2000:Q1 to 2024:Q4. Heteroskedasticity consistent 90 percent confidence intervals are reported using shaded horizontal bars.

Figure A.2: Relationship between Shortages and Inflation and Activity 8-Quarter Ahead



Note: The figure reports standardized coefficients of predictive regressions of inflation and economic activity on the shortage index estimated on quarterly data. The dependent variable for each regression is the log difference between $t+8$ and t of the variable listed in each row (the price deflator for inflation and real quantities for growth). Each regression includes as controls the quarterly changes of the dependent variable and the associated economic indicator or price deflator, contemporaneously and with three lags. Data are quarterly. The full sample runs from 1950:Q1 to 2024:Q4. We also partition the sample into two periods: 1950:Q1 to 1999:Q4, and 2000:Q1 to 2024:Q4. Heteroskedasticity consistent 90 percent confidence intervals are reported using shaded horizontal bars.

Figure A.3: Effects of Shortages on Inflation and Economic Activity 4-Quarter Ahead, with and without controls



Note: The figure reports standardized coefficients of predictive regressions of inflation and economic activity on the shortage index estimated on quarterly data. The dependent variable for each regression is the log difference between $t+4$ and t of the variable listed in each row (the price deflator for inflation and real quantities for growth). Each row in the Figure shows how the standardized coefficients change with the inclusion of the controls listed. The full sample runs from 1950:Q1 to 2024:Q4. We also partition the sample into two periods: 1950:Q1 to 1999:Q4, and 2000:Q1 to 2024:Q4. The regressions controlling for inflation expectations for the earlier sample start in 1982:Q1. Inflation expectations are from the Federal Reserve Bank of Cleveland via FRED. Heteroskedasticity consistent 90 percent confidence intervals are reported using shaded horizontal bars.